CISCO CCNA CRAM GUIDE V6.0

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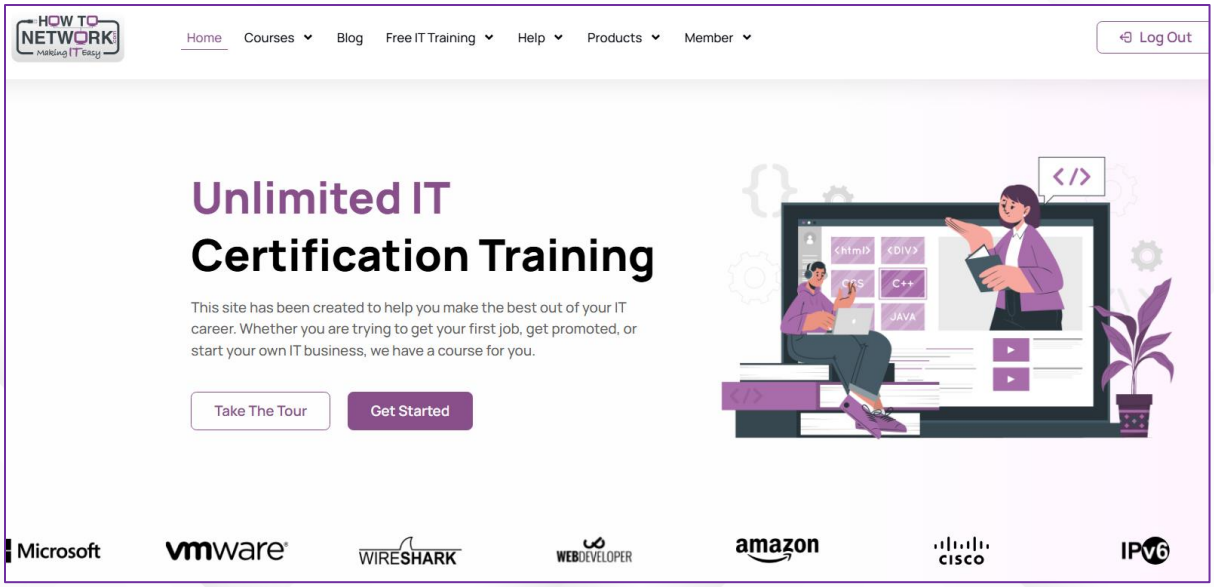
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Welcome

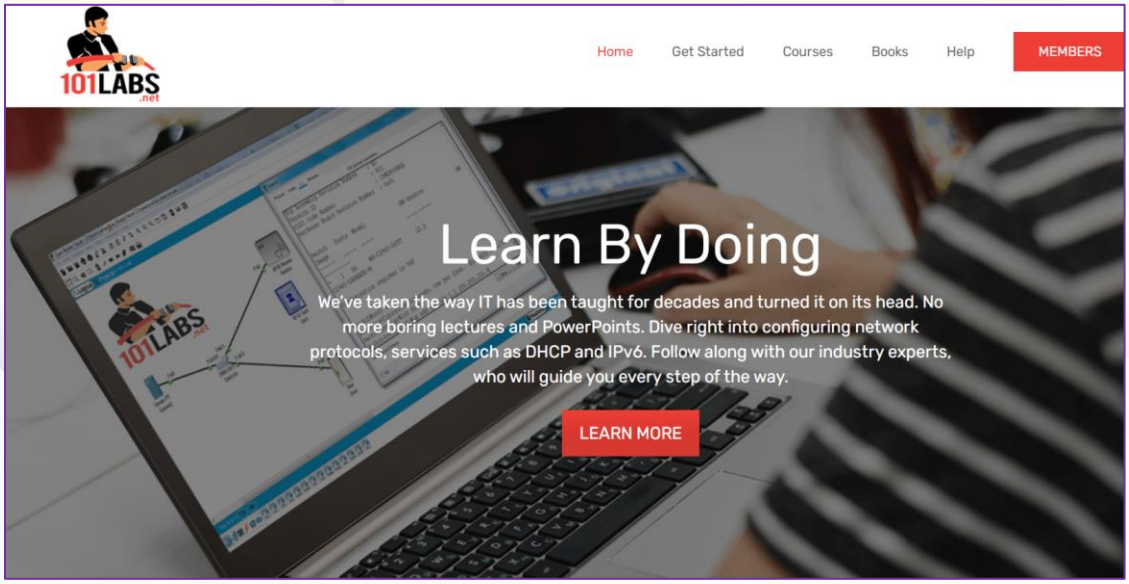
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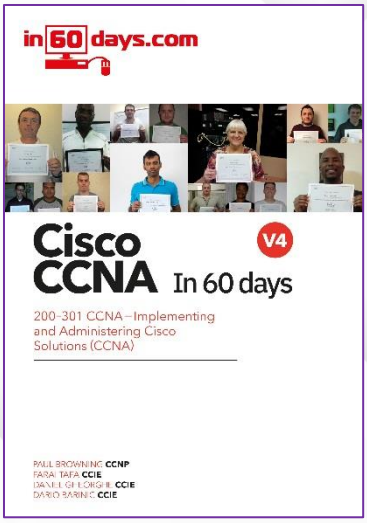
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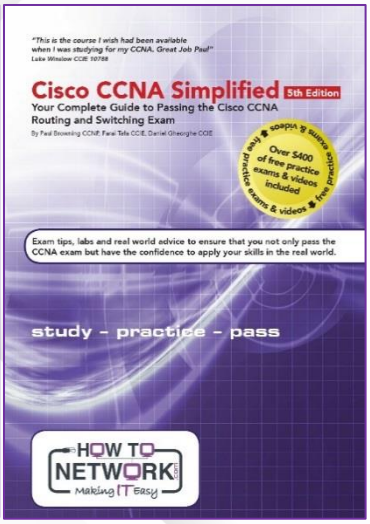


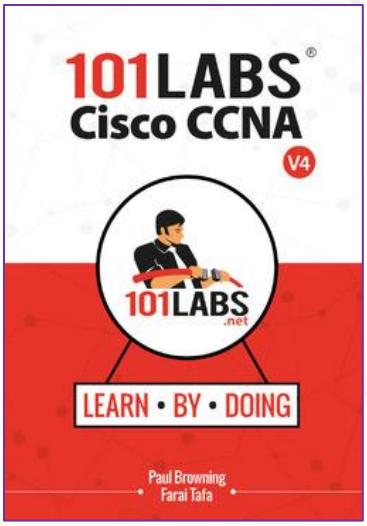
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Introduction

The complete contents of this guide MUST be committed to memory before attempting the exam. This cram guide is NOT a brain dump, so we have no way of knowing what exactly you are going to be asked about in the exam. We have taken the CCNA exam syllabus and extracted all the key points we think may come up for each syllabus topic.

Cisco DOES test you on other topics NOT listed in the exam syllabus. It's very unfair but it's out of our hands, so if you see something in this guide which appears to not belong, please bear that in mind.

In order to pass the CCNA exam, you must study theory daily, memorize this cram guide, take daily practice exams AND do hands-on labs. You must also drill subnetting every day until you know it back to front and inside out and are super-fast.

The only way to remember everything in this guide is to read it over and over and then write it out from memory, without looking. Eventually, it will sink in.

1.0 Network Fundamentals

Topologies

There are differences between the physical layout and how the topology works logically.

Topology	Physical	Logical
Bus	Bus	Bus
Star	Star	Bus
Token Ring	Star	Ring
Point-to-Point	Bus	Bus
FDDI	Ring	Ring

Firewalls

Firewalls filter traffic into and out of the network, and they are usually positioned at the network entry point or between critical modules. Most firewalls work at Layer 4, which is based on TCP or UDP port numbers (usually called traditional or legacy firewalls), although some can operate at Layer 7 (usually called **next-generation firewalls**).

First generation firewalls provided basic filtering capabilities at Layer 3 and Layer 4 of the OSI Model. Second-generation monitored traffic at Layer 3, Layer 4, and Layer 5 of the OSI Model. Third-generation firewalls provided firewalling capabilities at Layer 3, Layer 4, Layer 5, and Layer 7 of the OSI Model. And, finally, fourth-generation firewalls operate at Layer 3, Layer 4, Layer 5, and Layer 7 of the OSI Model and use a concept referred to as Stateful inspection.

A stateful firewall can track, for example, the number of connections to a Web server per second to understand what is typical (e.g. five to six connections per second). Then, if the state changes, that is, if the connections go to 100 per second, the stateful firewall will notice this and then analyze what is happening.

A stateless firewall doesn't have the enhanced capability to make decisions based on historical data (i.e. it isn't aware of whether the packet is part of an existing connection

or a new one). Stateless firewalls watch real-time network traffic and block traffic based on static values, such as source or destination address. They do not take into account traffic patterns or the history of data flows.

Intrusion Detection System (IDS) and Intrusion Prevention System (IPS)

IDS and IPS devices function by performing traffic inspection to detect unauthorized traffic that tries to enter the enterprise network. Their main role is to monitor networks for intrusions or other malicious activity. The actions taken by a device in a promiscuous mode include sending alerts, alarms, log messages, or SNMP traps.

The major difference between IPS and IDS is that IPS devices operate in line with the traffic (meaning they are placed in the middle of the traffic flow and all the packets pass through the inspection device), while IDS devices only retrieve a copy of the traffic so they can analyze it.

Routers

A router is designed to store a directory of networks. Rather than concerning itself with which PC is where, a router's job is to find out where different networks are. It then sends the traffic via the best path.

Switches

Switches operate by building a list of which PCs are connected to which ports, allowing the available bandwidth to be used a lot more efficiently. If a PC wants to send data to another PC via a switch, the switch will forward the traffic only to the port to which the intended recipient is connected. If it doesn't know the port, it will send out a broadcast to find out where in the network the PC is.

Layer 2 switches operate at the OSI Data Link layer using MAC addresses for fast intra-VLAN switching within a single broadcast domain, but they cannot route traffic between VLANs without an external router.

Layer 3 switches combine Layer 2 functionality with Network layer routing via IP addresses, enabling efficient inter-VLAN communication, ACLs, and protocols like OSPF, making them essential for scalable enterprise networks in CCNA scenarios.

Wireless Networking

Wireless networks use carrier sense multiple access with collision avoidance (CSMA/CA) to prevent frames from colliding. The centralized administration hub is called a Wireless LAN Controller (WLC).

Access points (APs) are used as hub devices to transport the signal across the network. CSMA/CD has been replaced by Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA). Wireless LANs have unique Service Set IDs (SSIDs), which can be seen at home or a local coffee shop when you click "scan" on your mobile phone and are asked for the password to connect to a network.

With WLCs, the APs no longer act as individual devices that store information on endpoints; rather, they become "lightweight" (morphing from APs to lightweight WAPs, or LWAPs) and forward everything they learn, such as WLAN (SSID) information, roaming, or authentication, to the centralized WLC.

Multiple APs servicing the same SSID are referred to as an Extended Service Set (ESS). A single AP is referred to as a Basic Service Set (BSS).

Power Over Ethernet (PoE)

Power over Ethernet (PoE) is a technology that allows switches to pass electrical power along with data on Ethernet cables. This is usually done on the four extra wires in UTP cables, the ones that are not used for data transmission.

PoE is generally used to power up non-critical network devices in order to reduce cabling necessities in the office. Such devices include:

- IP phones
- Video conferencing cameras
- Surveillance cameras
- Wireless access points

PoE is supported only on special switch models, which are usually more expensive.

OSI Model

Layer	Encapsulation	Function	Services	Device
7. Application	Data	Establishes availability of resources	FTP, SMTP, Telnet, POP3	
6. Presentation	Data	Compression, encryption and decryption	JPEG, GIF, MPEG, ASCII	
5. Session	Data	Establishes, maintains and terminates sessions	NFS, SQL, RPC	
4. Transport	Segment	Establishes end-to-end connection. Uses virtual circuits, buffering, windowing and flow control	TCP, UDP, SPX	
3. Network	Packet	Determines best path for packets to take.	IP, IPX	Router
2. Data Link (LLC MAC)	Frame	Transports data across a physical connection. Error detection	Frame Relay, PPP, HDLC	Switch/Bridge
1. Physical	Bits	Puts data onto the wire		Hub/Repeater /Concentrator /MAU

Layers - All People Seem To Need Data Processing. Data formats - Don't Some People Fry Bacon.

7. Application Layer

This provides services to lower layers; it enables program-to-program communication and determines if sufficient resources exist for communication. Examples are e-mail gateways (SMTP), TFTP, FTP and SNMP.

6. Presentation Layer

This presents information to the Application Layer. Compression, data conversion, encryption and standard formatting occur here. Contains data formats JPEG, MPEG, MIDI, TIFF.

5. Session Layer

This establishes and maintains communication 'sessions' between applications (dialogue control). Sessions can be simplex (one direction only), half-duplex (one direction at a time) or full duplex (both ways simultaneously). Session layer keeps different application data separate from other applications. Protocols include NFS, SQL, X Window, RPC, ASP, and NetBIOS Names.

4. Transport Layer

This is responsible for the end-to-end integrity of data transmission and establishes a logical connection between sending and receiving hosts via 'virtual circuits'. Windowing works at this level to control how much information is transferred before an acknowledgement is required. Data is segmented and reassembled at this layer. Port numbers are used to keep track of different conversations crossing the network at the same time. It supports TCP. UDP, SPX, NBP; segmentation works here (Segments) and error correction (not detection).

3. Network Layer

This routes data from one node to another and determines the best path to take. Routers operate at this level. Network addresses are used here which are used for

routing (Packets). Routing tables, subnetting and control of network congestion occur here. Routing protocols, regardless of which protocol they run over, reside here. IP, IPX, ARP, IGRP, Appletalk.

2. Data Link Layer

This is sometimes referred to as the LAN layer. It is responsible for the physical transmission of data from one node to another. Error detection occurs here. Packets are translated into frames here and hardware address is added. Bridges and switches operate at this layer.

Logical Link Control sub layer (LLC) 802.2: manages communications between devices over a single link on a network. It uses Service Access Points (SAPs) to help lower layers talk to the Network Layer.

Media Access Control (MAC) 802.3: builds frames from the 1's and 0's that the Physical Layer (address = 6-byte/48 bit) picks up from the wire as a digital signal and runs a Cyclic Redundancy Check (CRC) to assure no bits were lost or corrupted.

1. Physical Layer

This puts data onto the wire and takes it off, physical layer specifications such as the connectors, voltage, physical data rates and DTE/DCE interfaces. Some common implementations include Ethernet/IEEE 802.3, Fast Ethernet, and Token Ring/IEEE 802.5.

Cisco Hierarchical Model

Cisco has taken a modular approach to the construction of their network. The main advantage of this approach is that each module can be built by specialist teams (e.g., security, voice, routing, etc.) and supervised by the network architect. This (according to Cisco) is the key to simplifying the network.

There are many benefits to using the Cisco network design methodology, and it is worth making a note of them in case they appear as options in an exam question:

- Ease of understanding and implementation

- Flexibility
- Cost savings
- Modularity
- Easily modified
- Easy network growth
- Facilitates summarization (small routing tables and network stability)
- Fault isolation

In the Cisco Hierarchical Model, the layers of the network are broken down into the following:

- Core (backbone) layer
- Distribution layer
- Access layer

These layers are illustrated below:

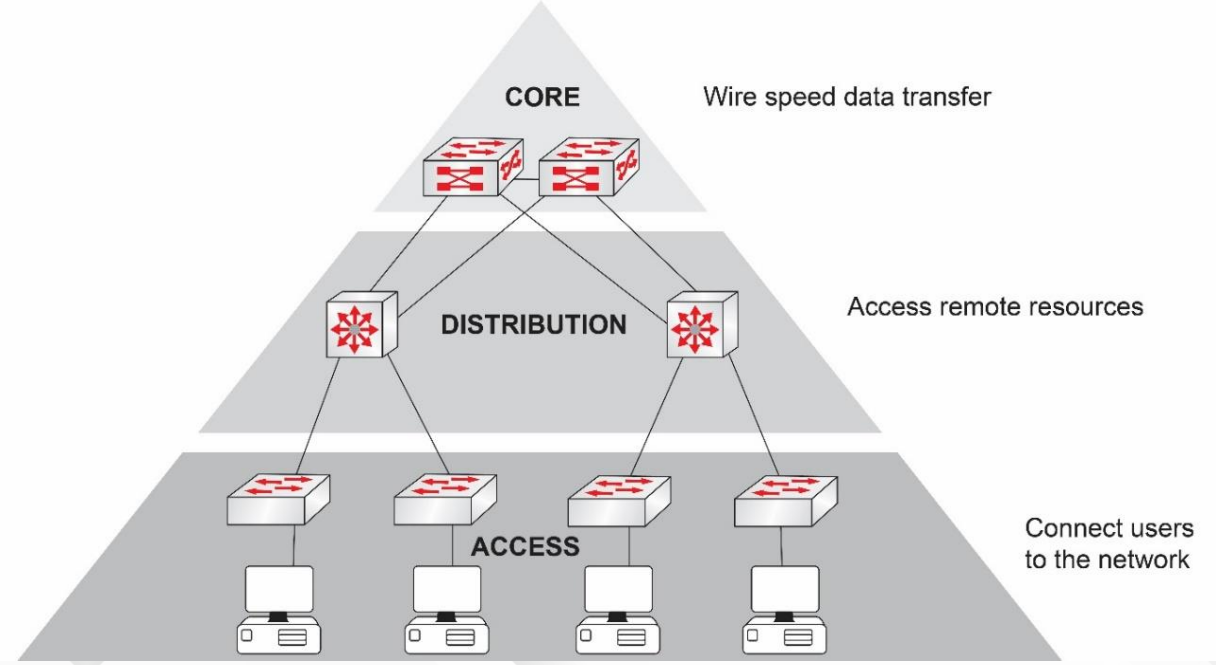


FIG – The layers of the Cisco Hierarchical Model

Collapsing the Layers

For any number of reasons, companies may not want to use the three-layer model because of cost issues, convenience, or the need to run a small network. In these instances, Cisco can simplify the three-layer model into a two-layer model.

This process is referred to as collapsing the design or collapsing the core, which only merges the core and distribution layers, leaving the access layer intact. Figure below shows three layers and a collapsed distribution/core layer:

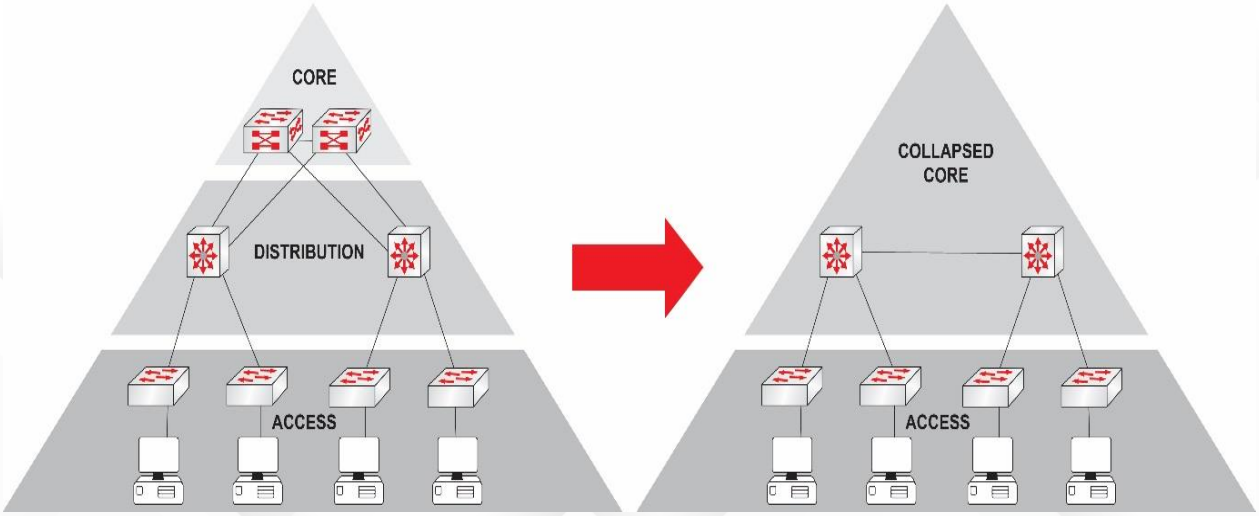


FIG – Three layers and a collapsed distribution/core layer

Core Layer – purpose is to switch traffic as quickly as possible. Fast transport to enterprise services (Internet, etc.). No packet manipulation, VLANs, access-lists. High-speed access required, such as FDDI, ATM.

Distribution Layer – time-sensitive manipulation such as routing, filtering and WAN access. Broadcast/Multicast, media translations, security.

Access Layer – switches and routers, segmentation occurs here and workgroup access. Static (not dynamic) routing.

Spine-leaf architecture is usually deployed as two layers: spines (such as an aggregation layer), and leaves (such as an access layer). Spine-leaf topologies provide high-bandwidth, low-latency, nonblocking server-to-server connectivity.

Spine-Leaf Network Topology

Also referred to as Leaf-Spine, the Spine-Leaf design is aimed at data centers that need to quickly move data on a massive scale. The three-tier model cannot serve this need and so is being replaced to match evolving needs.

Devices in a spine-leaf design are always exactly the same number of segments away and provide consistent latency for traffic. The two-layer design allows for this with the Leaf layer consisting of access switches that provide network access to servers, firewalls, and edge routers. The Spine layer is made up of layer 3 switches and are the backbone of the network.

Every Leaf switch is connected to every Spine switch in a full-mesh, and dynamic routing is used to interconnect the two layers. This type of data center design supports an East-West data flow where traffic is designed to travel inside the data center and not outside to an ISP or another network.

An advantage of using routing as opposed to layer 2 switching is that all links can be used and loops are avoided whereas a Spanning Tree Protocol (STP) would necessarily involve closing links to traffic in order to prevent loops. Should a Spine switch fail for any reason, traffic performance would only be slightly degraded.

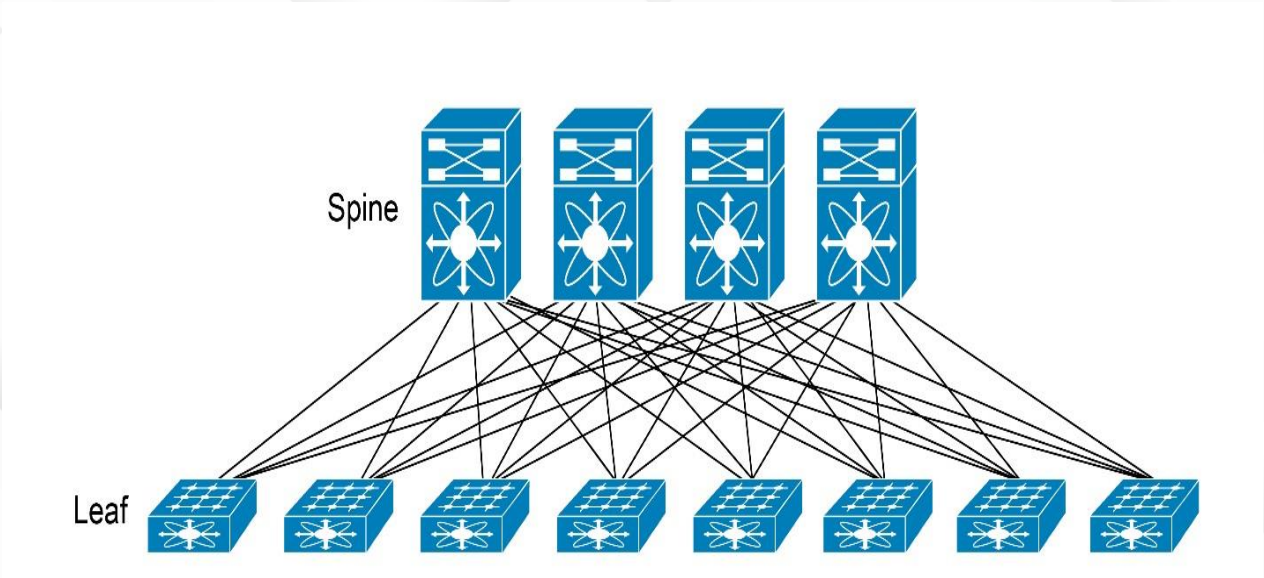


FIG – Spine-leaf network topology

TCP/IP

Port Numbers

These are used to connect to various services and applications and piggyback onto IP addresses. Common port numbers you should know include:

20	File Transfer Protocol – Data (TCP)
21	File Transfer Protocol – Control (TCP) (Listens on this port)
22	SSH (TCP)
23	Telnet (TCP)
25	Simple Mail Transfer Protocol (TCP)
53	Domain Name Service (TCP/UDP)
69	Trivial File Transfer Protocol (UDP)
80	HTTP/WWW (TCP)
110	Post Office Protocol 3 (TCP)
119	Network News Transfer Protocol (TCP)
123	Network Time Protocol (UDP)
161/162	Simple Network Management Protocol (UDP)
443	HTTP over Secure Sockets Layer (HTTPS) (TCP)
TCP	(protocol 6) reliable, sequenced connection-oriented delivery, 20-byte header.
UDP	(protocol 17) connectionless, unsequenced best effort delivery, 8-byte header. Sends data but does not check to see if it is received.
Telnet	used to connect to a remote device (TCP). A password and username are required to connect. Telnet tests all seven layers of the OSI model.
FTP	Connection orientated (TCP) protocol used to transfer large files.
TFTP	connectionless (UDP) protocol used for file transfer.
SNMP	allows remote management of network devices.
ICMP	supports packets containing error, control and informational messages. Ping uses ICMP to test network connectivity. Traceroute also uses ICMP. As with ping, Cisco routers allow you to perform an extended traceroute command that can define a number of associated parameters, most of them similar to the ping-related parameters:

```
Router#traceroute
Protocol [ip]:
Target IP address: 192.168.1.1
Source address: 10.10.10.2
Numeric display [n]:
Timeout in seconds [3]:
Probe count [3]:
Minimum Time to Live [1]:
Maximum Time to Live [30]:
Port Number [33434]:
Loose, Strict, Record, Timestamp, Verbose[none]:
Type escape sequence to abort.
Tracing the route to 192.168.1.1
1 10.10.10.1 76 msec *  56 msec
```

ARP – used to map an IP address to a physical (MAC) address. A host wishing to obtain a physical address broadcasts an ARP request onto the TCP/IP network. The host replies with its physical address. The dash below means that the address is local to the device. Otherwise, it will show an age in minutes.

```
Router#show arp
Protocol  Address  Age (min)  Hardware Addr   Type   Interface
Internet  192.168.1.1     -   0002.4A4C.6801  ARPA   FastEthernet0/0
```

DNS – resolves hostnames to IP addresses (not the other way around). To configure the router to use a host on the network, use the command `ROUTER(config)#ip name-server 4.2.2.2` and to configure DNS the command `ip name-server` is usually already turned on for the router configuration by default. If you want hosts on the network to use the router as a proxy DNS server, put in the command `ROUTER(config)#ip dns server` onto the router.

DHCP – involves a central server or devices which relay TCP information to hosts on a network. You can configure a router to be a DHCP server with the below configuration. You must have hosts on the same LAN as the router interface:

```
Router(config)#ip dhcp pool E00_DHCP_Pool
Router(dhcp-config)#network 10.10.10.0 255.255.255.0
Router(dhcp-config)#dns-server 24.196.64.39 24.196.64.40
Router(dhcp-config)#domain-name mydomain.com
Router(dhcp-config)#default-router 10.10.10.254
Router(dhcp-config)#lease 1
Router(dhcp-config)#exit
Router(config)#ip dhcp excluded-address 10.10.10.1
```

Routers will not forward DHCP broadcasts, so if your DHCP server is on another subnet, add the command ip helper-address [ip address].This is referred to as dhcp relay.

NTP – a protocol that is designed to time-synchronize a network of machines. NTP is documented in RFC 1305 and runs over UDP.

```
R2(config)#ntp server 10.0.0.1

R2#show ntp associations
 address      ref clock     st   when   poll reach   delay   offset   disp
*~10.0.0.1   127.127.7.1   5    44     64   377     3.2     2.39     1.2
* master (synced), # master (unsynced), + selected, - candidate, ~
configured
```

If you wanted to set the router as the NTP server, you would add the below command and (optionally) a stratum level.

```
R1(config)#ntp master?
  <1-15> Stratum number
```

Cisco IOS

Six modes

User EXEC: - Router>
Privileged EXEC: - `Router#`
Global Configuration: - `Router(config)#`
ROM Monitor: - > or `rommon>`
Setup: - series of questions
RXBoot:- `Router<boot>`

Editing Commands

Ctrl+W - Erases a word
Ctrl+U - Erases a line
Ctrl+A - Moves cursor to beginning of line
Ctrl+E - Moves cursor to end of line
Ctrl+F - (or right arrow) – Move forward one character
Ctrl+B - (or left arrow) – Move back one character
Ctrl+P - (or up arrow) – Recalls previous commands from buffer
Ctrl+N - (or down arrow) – Return to more recent commands in buffer

Esc+B - Move back one word
Esc+F - Move forward one word
Tab - completes a command you have started (see below)

```
Router#copy ru ← press tab key after the 'u'
Router#copy running-configuration
```

? gives you the command options

```
Router#copy ?
  flash:          Copy from flash: file system
  ftp:            Copy from ftp: file system
  nvram:          Copy from nvram: file system
  running-config  Copy from current system configuration
  startup-config  Copy from startup configuration
  system:         Copy from system: file system
  tftp:           Copy from tftp: file system (truncated to save space)

or the commands beginning with the letters you have typed:

Router#a?
access-enable access-profile  access-template
```

Cabling

	Hub	Switch	Router	Workstation
Hub	Crossover	Crossover	Straight	Straight
Switch	Crossover	Crossover	Straight	Straight
Router	Straight	Straight	Crossover	Crossover
Workstation	Straight	Straight	Crossover	Crossover

Pinouts

Crossover Pinouts

1 → 3
2 → 6
3 → 1
6 → 2

Straight Through

1 → 1
2 → 2
3 → 3
4 → 4
5 → 5
6 → 6
7 → 7
8 → 8

Rollover (PC to Console/Aux port)

1 → 8
2 → 7
3 → 6
4 → 5
5 → 4
6 → 3
7 → 2
8 → 1

Two types of crosstalk can occur on twisted pair cables:

Near end crosstalk (NEXT) and Far end crosstalk (FEXT)

You can change the interface speed and duplex settings which are usually set to auto (and this can cause issues).

```
Router#show interface FastEthernet0
FastEthernet0 is up, line protocol is up
  Hardware is DEC21140AD, address is 00e0.1e3e.c179 (bia 00e0.1e3e.c179)
  Internet address is 1.17.30.4/16
  MTU 1500 bytes, BW 10000 Kbit, DLY 1000 usec, rely 255/255, load 1/255
  Encapsulation ARPA, Loopback not set, keepalive set (10 sec)
```

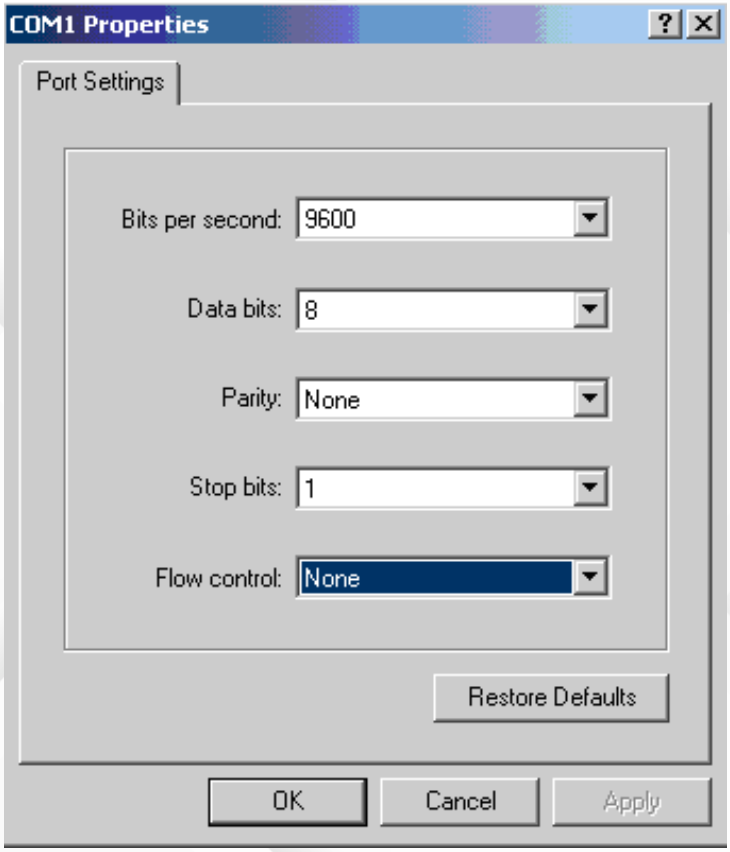
```
  Half-duplex, 10Mb/s, 100BaseTX/FX

Router(config)#int f0
Router(config-if)#duplex ?
  auto  Enable AUTO duplex configuration
  full  Force full duplex operation
  half  Force half-duplex operation

Switch(config-if)#duplex full

Router#config t
Router(config)#interface g0/0
Router(config-if)#speed ?
  10    Force 10 Mbps operation
  100   Force 100 Mbps operation
  1000  Force 1000 Mbps operation
  auto  Enable AUTO speed configuration
```

If you need to console to a router, here are the settings:



CDP / LLDP

Cisco Discovery Protocol runs only on Cisco devices (proprietary); it allows you to gather information about other routers and switches. It is enabled by default.

`Router#show cdp neighbors` (note: Cisco uses US spelling conventions)

This command displays the neighboring router or switches hostname, hardware platform, port identifier and capabilities list.

```
Router#show cdp neighbors detail
```

This command displays more detail than the previous one. You can view IP address, IOS release and duplex setting.

To turn CDP off an interface, use the command:

```
Router(config-if)#no cdp enable
```

To turn CDP off on your entire router or switch, use the command:

```
Router(config)#no cdp run
```

LLDP is an open standard under IEEE standard 802.1AB and shares many of the same commands and outputs as CDP. It's disabled by default.

```
Switch#configure terminal
Switch(config)#lldp run
Switch(config)#end
```

You can also enable or disable per interface:

```
Switch(config)#interface GigabitEthernet1/1
Switch(config-if)#lldp transmit
Switch(config-if)#lldp receive
Switch(config-if)#end
```

Hex Numbering

You know that decimal numbers consist of 10 digits ranging from 0 to 9. Binary consists of two digits ranging from 0 to 1. Hex numbering ranges from 0 to F and has 16 digits. These addresses are also referred to as base 10, base 2, and base 16, respectively. You can see that each numbering system starts with a zero, so:

Decimal – 0123456789
Binary – 01
Hex – 0123456789ABCDEF

When you write these addresses, you may not realize it, but you are using columns from right to left; the rightmost is the one column and the next column is the base number times the preceding column, so:

Numbering Base	N to 3rd power	N to 2nd power	N to 1st power	N
10 – Decimal	1000	100	10	1
2 – Binary	8	4	2	1
16 – Hex	4096	256	16	1

You can see that each successive column from the right increases in value. For decimal numbering, it is 10 multiplied by 1. For binary, it is 1 and then 1 multiplied by the numbering system of 2. If you compare the three numbering systems up to the last hex digit, you can begin to see why hex is the preferred format for IPv6 addressing.

Decimal	Binary	Hex
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8

Decimal	Binary	Hex
9	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

IP Addressing and Subnetting

Originally, a range of IP addresses was reserved for use by any person or company who needed them. This was to save IP addressing space. They can't be routed on the Internet, so you would need a public address and NAT. RFC 1918 private address ranges:

- 192.168.0.0 - 192.168.255.255
- 172.16.0.0 - 172.31.255.255
- 10.0.0.0 - 10.255.255.255

IP address 127.0.0.1 is reserved for loopback testing on interfaces to check if the TCP stack is working.

IP addresses were put into classes, but this is no longer relevant due to VLSM. Here are the various classes along with the default subnet mask and number of networks and hosts per network.

Class	Format / Default Mask	Leading Bit Pattern	Network Address Range	Max Networks	Max hosts/nodes
A	N.H.H.H 255.0.0.0	0	0-126	126	16,777,214
B	N.N.H.H 255.255.0.0	10	128-191	16,384	65,534
C	N.N.N.H 255.255.255.0	110	192-223	2,097,152	254
D	n/a	1110	224-239	multicast	n/a
E	n/a	11110	240-255	experimental	n/a

Network number 127 reserved for loopback testing (127.0.0.1 local loopback).

- **Unicast data:** data sent to one device
- **Multicast data:** data sent to a group of devices
- **Broadcast data:** data sent to all devices

- **Collision domain:** includes all devices that share the same bandwidth; collision domains are separated by switch ports
- **Broadcast domain:** includes all devices that receive broadcast messages; broadcast domains are separated by routers (and VLANs)

Subnetting

Max # of Subnets = 2(to the power of masked bits)
Max # of Hosts (per subnet) = 2(to the power of unmasked bits)

Easy Subnetting

What network is host 172.16.5.68 255.255.255.240 on?

256-240 = 16 - so you have the subnets going up in increments of 16 starting with zero. Each subnet will need to have a subnet and a broadcast number, so this leaves 14 usable host numbers per subnet. The subnets start at 0,16,32,48, 64, 80….224, 240. We stop at 240 because this is the last number of the above subnet mask.

	Subnet	First host	Last host	Broadcast
1.	172.16.5.0	1	14	15
2.	172.16.5.16	17	30	31
3.	172.16.5.32	33	62	63
4.	172.16.5.64*	65	78	79
Last	172.16.255.240	241	254	255

Looking at the 172.16.5.68 host address, we are clearly looking to find the host on the fourth octet because this is where the non-zero mask is (255.255.255.240). We just keep adding 16 until we find the range that the host number 68 is in. In this case, 172.16.5.64 is indicated by the asterisk*.

The Super Subnetting Chart™ will also provide the answers for you very quickly. To get to the 240 mask, we tick down four subnet numbers and then tick four along the top to get the subnet increment (i.e. 16). I know 172 is a class C network, but this doesn't matter for subnetting. We just work on the octet the subnet is happening in.

	Bits	128	64	32	16	8	4	2	1
Subnets		✓	✓	✓	✓				
128	✓								
192	✓								
224	✓								
240	✓								
248									
252									
254									
255									
Powers of Two	**Subnets**	**Hosts -2**							
2									
4									
8									
16									
32									
64									
128									
256									

Super Subnetting Chart © Paul Browning 2005-2025

If you see a slash address such as 192.168.1.2 / 26 then you simply have to convert that 26 into a subnet mask. Using the Super Subnetting Chart is the easiest way or you can simply count up in octets remembering that 255 is 8 in binary bits. We are looking to reach 26 binary bits. We know that 255.255.255.0 is 8+8+8 binary bits, giving us 24. To get to 26, we need to add another two binary bits which are a tick in the 128 and 192 (128 + 64 = 192) binary columns.

To work out the subnet increment, simply tick two along the top column (or take 192 away from 256).

	Bits	128	64	32	16	8	4	2	1
Subnets		✓	✓						
128	✓								
192	✓								
224									
240									
248									
252									
254									
255									
Powers of Two	**Subnets**	**Hosts -2**							
2									
4									
8									
16									
32									
64									
128									

Super Subnetting Chart © Paul Browning 2005 - 2025

128+64 gives you 192 so the /26 mask gives you 255.255.255.192 which is 26 binary bits. You can tick two across the top to reveal the subnet increments (i.e. multiples of 64).

Working out how many subnets is also a simple process.

If you have the IP address 192.168.1.0 with the default /24 mask and the client wants five subnets creating out of that, you simply tick down the powers of two (Subnets) numbers until you get the required amount of subnets.

	Bits	128	64	32	16	8	4	2	1
Subnets									
128	✓								
192	✓								
224	✓								
240									
248									
252									
254									
255									
Powers of Two	**Subnets**	**Hosts -2**							
2	✓	✓							
4	✓	✓							
8	✓	✓							
16		✓							
32		✓							
64									
128									

Super Subnetting Chart © Paul Browning 2005 - 2025

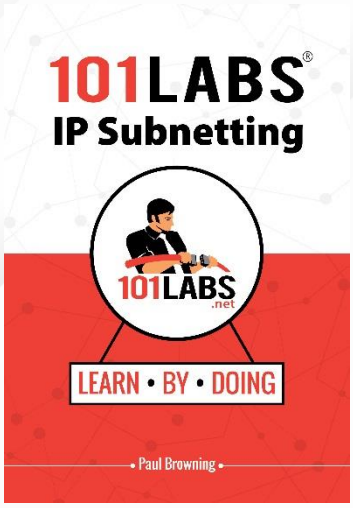
Ticking down three columns gives you eight. I know that the requirement is five but this is the closest you can get for the client. Taking three bits from the hosts leaves five remaining host bits (8–3 = 5). Tick down five boxes in the host column and take two away (for the subnet and broadcast) to give you 30 hosts per subnet. Voila!

If you want to know the subnet mask, tick down three (for the subnet bits stolen) columns in the top subnet column giving you 224 or in full 255.255.255.224 which is 8+8+8+3 or /27 masked bits.

I break subnetting down from complete novice to expert level on my IP Subnetting – Zero to Guru course on howtonetwork.com or you can grab the book from Amazon.



If you just need to practice subnetting, VLSM and route summarization, then please check out my 101 Labs – IP Subnetting book on Amazon.



Route Summarization

You need to be able to work this out for the exam. It is basically advertising out as few routes as possible from your network. You can only really work out a summary route by converting the IP address into binary (sorry). If you don't do this, then you have no way of knowing if you are advertising the correct summary route, which will lead to problems on your network.

Firstly, write out all of the network addresses in full, and then the binary versions to the right of that.

172.16.8.0	*10101100.00010000.00001*000.00000000
172.16.9.0	*10101100.00010000.00001*001.00000000
172.16.10.0	*10101100.00010000.00001*010.00000000
172.16.11.0	*10101100.00010000.00001*011.00000000
172.16.12.0	*10101100.00010000.00001*100.00000000
172.16.13.0	*10101100.00010000.00001*101.00000000
172.16.14.0	*10101100.00010000.00001*110.00000000
172.16.15.0	*10101100.00010000.00001*111.00000000
Matching Bits	*10101100.00010000.00001* = 21 bits

I have italicized the bits in each address which match. You can see that the first 21 bits match on every address so your summarized route can reflect these 21 bits:

172.16.8.0 255.255.248.0

VLSM

VLSM principles will let you take a network and slice it down into smaller chunks. Those chunks can then be sliced into smaller chunks and so on. You only reach the limit when you get to the mask 255.255.255.252 or /30 because this gives you two usable hosts which are the minimum you would need for any network.

Let's take network 200.100.100.0 /24

You change the mask from /24 to /25 so this happens:

Original mask (last octet)	00000000	1 Subnet	254 hosts
New mask (subnet 1)	**0**0000000	200.100.100.0 - subnet 1	126 hosts
New mask (subnet 2)	**1**0000000	200.100.100.128 - subnet 2	126 hosts

Now, you have two subnets. If you take the new subnet 2 of 200.100.100.128 and break it down further by changing the mask from /25 to /26 you get this:

Original mask (last octet)	10000000	1 Subnet	126 hosts
New mask (subnet 1)	10000000	200.100.100.128 - subnet 1	62 hosts
New mask (subnet 2)	11000000	200.100.100.192 - subnet 2	62 hosts

The rule is that you can only use each IP address once, no matter what the subnet mask is.

IPv6

In order to provide enough addresses for our needs many years into the future, IPv6 has been designed to provide many trillions of available addresses. In order to do this, the numbering range has been expanded from 32 binary bits to 128 bits. Every 4 bits can be represented as one hex digit (as can be seen from the chart above). Logic then dictates that two hex digits will give us 8 bits, which is a single byte, or octet.

The 128 bits are broken down into eight sets of 16 bits each separated by a colon when written in full format. Every 4-hex bit can range from 0000 to FFFF, with F being the highest digit available in hex numbering:

0000	**0000**	**0000**	**0000**	**0000**	**0000**	**0000**	**0000**
to	to	to	to	to	to	to	to
FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF	FFFF

An IPv6 address consists of 128 bits represented in hexadecimal format separated into eight parts, e.g. EEDE:AC89:4323:5445:FE32:BB78:7856:2022. There are no broadcast packets, only anycast – multicast – unicast.

The three ways in which IPv6 addresses can be represented are as follows:

- The preferred or complete address representation or form
- Compressed representation
- IPv6 addresses with an embedded IPv4 address

The preferred representation for an IPv6 address is the longest format, also referred to as the complete form of an IPv6 address. This format represents all 32 hexadecimal characters that are used to form an IPv6 address. This is performed by writing the address as a series of eight 16-bit hexadecimal fields, separated by a colon (e.g., 3FFF:1234:ABCD:5678:020C:CEFF:FEA7:F3A0).

The compressed representation allows for IPv6 addresses to be compressed in one of two ways. The first method allows a double colon (::) to be used to compress consecutive zero values in a valid IPv6 address for successive 16-bit fields comprised of zeros or for leading zeros in the IPv6 address. When using this method, it is important to remember that the double colon can be used only once in an IPv6 address.

The second method of IPv6 compressed address representation is applicable to each 16-bit field and allows leading zeros to be omitted from the IPv6 address. When using this method, if every bit in the 16-bit field is set to 0, then one zero must be used to represent this field. In this case, not all of the zero values can be omitted.

Complete IPv6 Addresses Using Both Compressed Form Methods

Complete IPv6 Address Representation	Compressed IPv6 Address Representation
0000:0000:0000:0000:1a2b:000c:f123:4567	::1a2b:c:f123:4567
FEC0:0004:AB10:00CD:1234:0000:0000:6789	FEC0:4:AB10:CD:1234::6789
3FFF:0c00:0000:1010:1A2B:0000:0000:DE0F	3FFF:c00:0:1010:1A2B::DE0F
2001:0000:0000:1234:0000:5678:af23:00d5	2001::1234:0:5678:af23:d5

Cisco IOS supports IPv6 commands in version 12.2(2)T and later.

IPv6 Address Types

- Link-Local addresses
- Site-Local addresses (depreciated)
- Aggregate Global Unicast addresses
- Multicast addresses

- Anycast addresses
- Loopback addresses
- Unspecified addresses

IPv6 Link-Local addresses can be used only on the local link (i.e. a shared segment between devices) and are automatically assigned to each interface when IPv6 is enabled on that interface. These addresses are assigned from the Link-Local prefix FE80::/10.

Aggregate Global Unicast addresses are the IPv6 addresses used for generic IPv6 traffic, as well as for the IPv6 Internet. These are similar to the public addresses used in IPv4. Aggregate Global Unicast addresses for IPv6 are assigned by the Internet Assigned Numbers Authority (IANA) and fall within the IPv6 prefix 2000::/3. This allows for a range of Aggregate Global Unicast addresses from 2000 to 3FFF.

The Multicast addresses used in IPv6 are derived from the FF00::/8 IPv6 prefix. In IPv6, Multicast operates in a different manner than that of Multicast in IPv4. IP Multicast is used extensively in IPv6 and replaces IPv4 protocols, such as the Address Resolution Protocol (ARP).

Anycast can be described simply as one-to-nearest communication because the nearest common address, based on routing protocol metrics, will always be preferred by the local device. In IPv6, there is no specially allocated range for Anycast as Anycast addresses use Global Unicast, Site-Local, or even Link-Local addresses.

Loopback addresses in IPv6 are used in the same manner as in IPv4. Each device has one IPv6 Loopback address, which is comparable to the 127.0.0.1 Loopback address used in IPv4, and this address is used by the device itself. IPv6 Loopback addresses use the prefix ::1.

In IPv6 addressing, unspecified addresses are simply Unicast addresses that are not assigned to any interface. These addresses indicate the absence of an IPv6 address and are used for special purposes that include IPv6 DHCP and DAD. Unspecified addresses are represented by all 0 values in the IPv6 address and can be written using

the :: prefix. In the preferred format, these addresses are represented as 0000:0000:0000:0000:0000:0000:0000:0000.

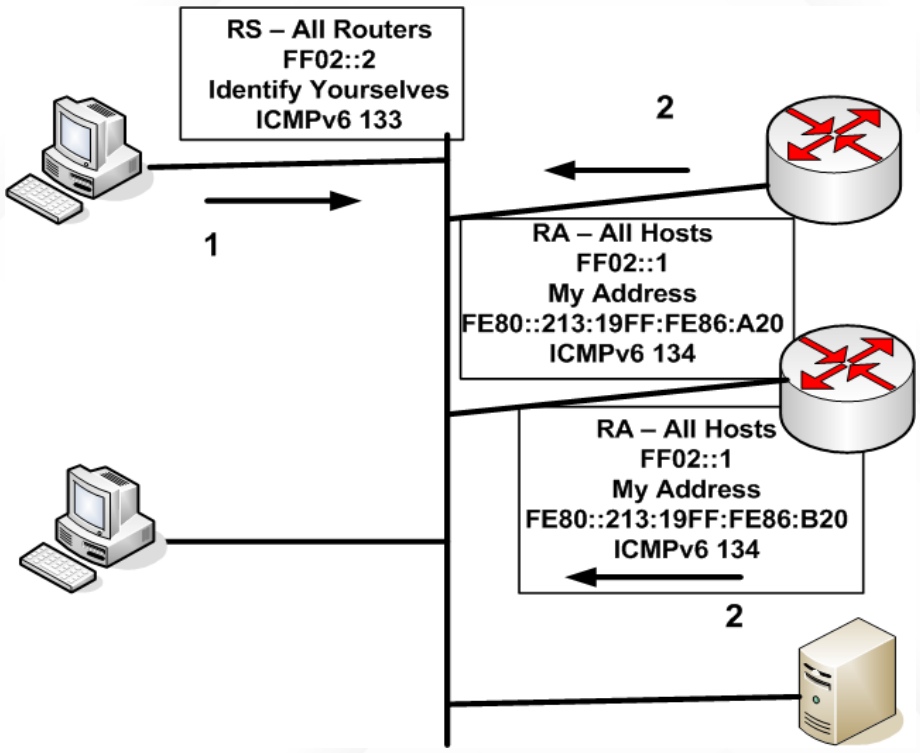
Neighbor Discovery Protocol

The IPv6 NDP enables the plug-and-play features of IPv6. It is defined in RFC 2461 and is an integral part of IPv6. NDP operates in the Link Layer and is responsible for the discovery of other nodes on the link, determining the Link Layer addresses of other nodes, finding available routers, and maintaining reachability information about the paths to other active neighbor nodes. NDP performs functions for IPv6 similar to the way ARP (which it replaces) and ICMP Router Discovery and Router Redirect Protocols do for IPv4.

Neighbor Discovery Protocol defines five types of ICMPv6 packets, which are listed and described below:

ICMPv6 Type	Message Type Description and IPv6 Usage
133	Used for Router Solicitation (RS) messages
134	Used for Router Advertisement (RA) messages
135	Used for Neighbor Solicitation (NS) messages
136	Used for Neighbor Advertisement (NA) messages
137	Used for Router Redirect messages

RS and RA messages are for router-to-host or host-to-router exchanges.



EUI Addressing

In the first step of creating the EUI-64 address, the value FFFE is inserted into the middle of the MAC address, thereby expanding the MAC (bia below) address from 48 bits, which is 12 hexadecimal characters, to 64 bits, which is 16 hexadecimal characters. The second step of EUI-64 addressing entails the setting of the seventh bit of the 64-bit address. This seventh bit is used to identify whether the MAC address is unique.

To flip the 7th bit (1-based indexing from the leftmost MSB) in the MAC address 0013.1986.0a20, start by converting the first octet (00) from hexadecimal to binary (00000000), where the 7th bit is 0; flipping it to 1 yields 00000110 in binary, which is 02 in hexadecimal. The resulting MAC address is 0213.1986.0a20.

```
R2#show interface FastEthernet0/0
FastEthernet0/0 is up, line protocol is up
  Hardware is AmdFE, address is 0013.1986.0a20 (bia 0013.1986.0a20)
   Internet address is 10.0.1.1/30
R2#show ipv6 interface FastEthernet0/0
FastEthernet0/0 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::213:19FF:FE86:A20
  Global unicast address(es):
    3FFF:1A2B:3C4D:5E6F:213:19FF:FE86:A20, subnet is
```

Verify IP Parameters for Client OS

Windows – use the cmd command to get the command line up and issue the ip config command or ipconfig /all for detail.

Linux - The command to display your IP information was ifconfig for many years. This command has been depreciated, and the new command is ip addr show.

MacOS - The GUI shows several options by default, choose the **System Settings** gear icon and then **Network**.



FIG 4.30 – Settings in Mac OS

Here, you will be able to see the basic IP information. If you choose **Options**, you can see more information, manually set your IP address, and change security.

Wan Protocols and Services

HDLC – Cisco defaults on serial WAN connections. No authentication available.

PPP – data link. Uses PAP (clear text) and CHAP (secure hash) authentication. Authentication is optional. Use PPP if connecting a Cisco router to a non-Cisco router.

```
Router(config)#hostname paul password cisco ← case sensitive
Router(config)#interface serial 0
Router(config-if)#encapsulation ppp
Router(config-if)#ppp authentication chap
```

Virtual Private Network

Virtual Private Network (VPN) is an encrypted network transported over a public network such as the Internet. VPN carries private data over a public network using advanced encryption and tunneling methods.

VPN provides:

- Privacy
- Authenticate
- Data integrity

VPN tunneling protocols:

- L2TP
- GRE
- IPSec
- L2F
- PPTP

There are many types of VPN technologies, including the following:

- Site-to-Site VPNs, or Intranet VPNs, for example Overlay VPNs (like Frame Relay) or Peer-to-Peer VPNs (like MPLS). You would use these when you want to connect different locations over the public infrastructure. When using a peer-to-peer infrastructure, you can seamlessly communicate between your sites without worrying about IP addressing overlap.
- Remote Access VPNs, for example Virtual Private Dialup Network (VPDN), which is a dial-up approach for the VPN that is usually done with security in mind.

Cloud Computing & Virtualization

Cloud computing and virtualization allow a company to quickly create a virtual server or entire network infrastructure (including routers, firewalls, and switches), scale up or down, install services and software, test it all, and then close it down, paying only for what was used.

Cloud computing can be employed as Software as a Service (SaaS), Platform as a Service (PaaS), or Infrastructure as a Service (IaaS).

SaaS is a type of cloud computing that provides answers to desktop needs for end-users, for example, Gmail from Google and Microsoft Office 365. PaaS enables developers to create and test software without investing in expensive hardware, for instance, Google App Engine. IaaS enables applications to run in the cloud instead of using their own infrastructure. Using IaaS, network administrators can build new servers and storage in a few minutes, for example, Amazon S3 storage.

Private cloud: In this type of model, organizations use their own hardware and software resources to achieve cloud services. This model mostly uses virtualization and operates entirely for the benefit of the organization.

Community cloud: This model consists of a pool of computer resources. These resources are available to different organizations with common needs. Clients can access the resources quickly and securely. In this model, clients are referred to as Tenants. Although clients are accessing the same computer resources, each client has their own computing environment with its own configuration and data. These details are stored securely.

Public cloud: This cloud model's services can be accessed by any Internet user. These computer resources are available to all public users. Examples of this cloud model are Gmail, Google Docs, and Hotmail. Currently, Amazon Web Services is the largest public cloud provider.

Hybrid cloud: This cloud model is a combination of both private and public clouds. Apart from the private cloud, users have no control of the hardware used or other

networking components. They can, however, control which OS is installed on the server as well as the applications.

Virtualization involves having a single physical device on top of which you use virtualization software that is able to separate virtual machines inside the physical device. The virtualization software will allocate a certain amount of disk space, memory, and CPU capacity to each virtual machine (VM) defined inside. If you wanted to build a new server, you would just carve out a new section of the physical device to create another virtual operating system and allocate the necessary resources, making it act (and think) exactly as if it were a physical device.

The software that makes this happen is called a virtual machine manager or a hypervisor (which is more than a supervisor). There are two types of hypervisors: type 1 and type 2.

The type 1 hypervisor is installed directly on the physical server (so it is called "bare metal" approach). Then instances of an operating system (OS) are installed on the hypervisor. The type 1 hypervisor has direct access to the hardware resources.

The type 2 hypervisor (or hosted hypervisor) runs on top of an operating system. Management console software is not required.

Server virtualization is a technology that uses hypervisors to abstract and partition physical server hardware into multiple isolated virtual machines (VMs), each running its own operating system and applications, to optimize resource utilization and scalability.

Containers are lightweight, portable virtualization units that package applications and their dependencies together to run consistently across different environments by sharing the host operating system's kernel, without the overhead of full VMs.

VRFs (Virtual Routing and Forwarding) are a Cisco networking technology that enables a single physical router to maintain multiple independent routing tables and forwarding instances, providing logical network segmentation for isolation and multi-tenancy without requiring separate hardware.

Wireless

With WLCs, the APs no longer act as individual devices that store information on endpoints; rather, they become "lightweight" (morphing from APs to lightweight WAPs, or LWAPs) and forward everything they learn, such as WLAN (SSID) information, roaming, or authentication, to the centralized WLC. The WLC then takes over the role of the AP by managing the wireless network with all of the information it learns from the LWAPs. LWAPs typically use a protocol called CAPWAP (Control and Provisioning of Wireless Access Points) to accomplish this task.

Every mobile device will join a wireless network identified by a Service Set ID (SSID). Once joined, the device will usually ignore other SSIDs. Multiple APs servicing the same SSID are referred to as an Extended Service Set (ESS). A single AP is referred to as a Basic Service Set (BSS).

Three main types of wireless frames are: the Data Frame, the Management Frame and the Control Frame. Association Response belongs to Management Frame. Association Response is sent in response to an association request.

WLANs support four levels of QoS:

Platinum -Voice
Gold-Video
Silver-Best Effort (default),
Bronze-Background.

WPA - Most WPA implementations use a pre-shared key (PSK), referred to as WPA Personal, and the Temporal Key Integrity Protocol (TKIP) for encryption. WPA was, in part, based on 802.11i, and it included 802.1x authentication (which we cover later).

WPA2 - WPA2 includes mandatory support for CCMP—an AES-based encryption mode. WPA2 is based around the superior AES CCMP algorithms because TKIP was depreciated. It should be obvious that WPA2 was meant as a replacement for WPA. WPA2 support is the minimum requirement for any device bearing the Wi-Fi trademark.

WPA3 - stronger encryption by AES with the Galois/Counter Mode Protocol (GCMP). It offers 192-bit cryptographic strength in WPA3-Enterprise mode. Protected Management Frames (PMFs) are used to secure management frames between the AP and client, to prevent hackers.

Troubleshooting

Always use a systematic and methodical approach to troubleshooting.

The first command to issue is show `ip interface brief` to establish if the interfaces are down or up. There are only a handful of ways to break any network in the exam.

Layer 1

Ensure that there is a clock rate on the DCE interface (use the `show controllers serial X` command to check what type of cable is attached – where X is the serial interface number).

Ensure that the no shut command has been applied to the interface.

Layer 2

Ensure that the correct encapsulation type is on the interface, i.e. HDLC, PPP, etc. (use the `show interface serial X` command to check). If it is not, then go into interface configuration mode and change it.

Layer 3

Ensure that the correct IP address and subnet mask is applied to the interface (commands above). Ensure that the correct networks are being advertised by the routing protocol (`show ip protocols`).

VLANs

For VLANs, ensure that the VLAN has been configured on the port and that the VLAN has been created and the interface is no shut if it's been shut. Check that there is a trunk link between the two switches and that no VLANs have been blocked.

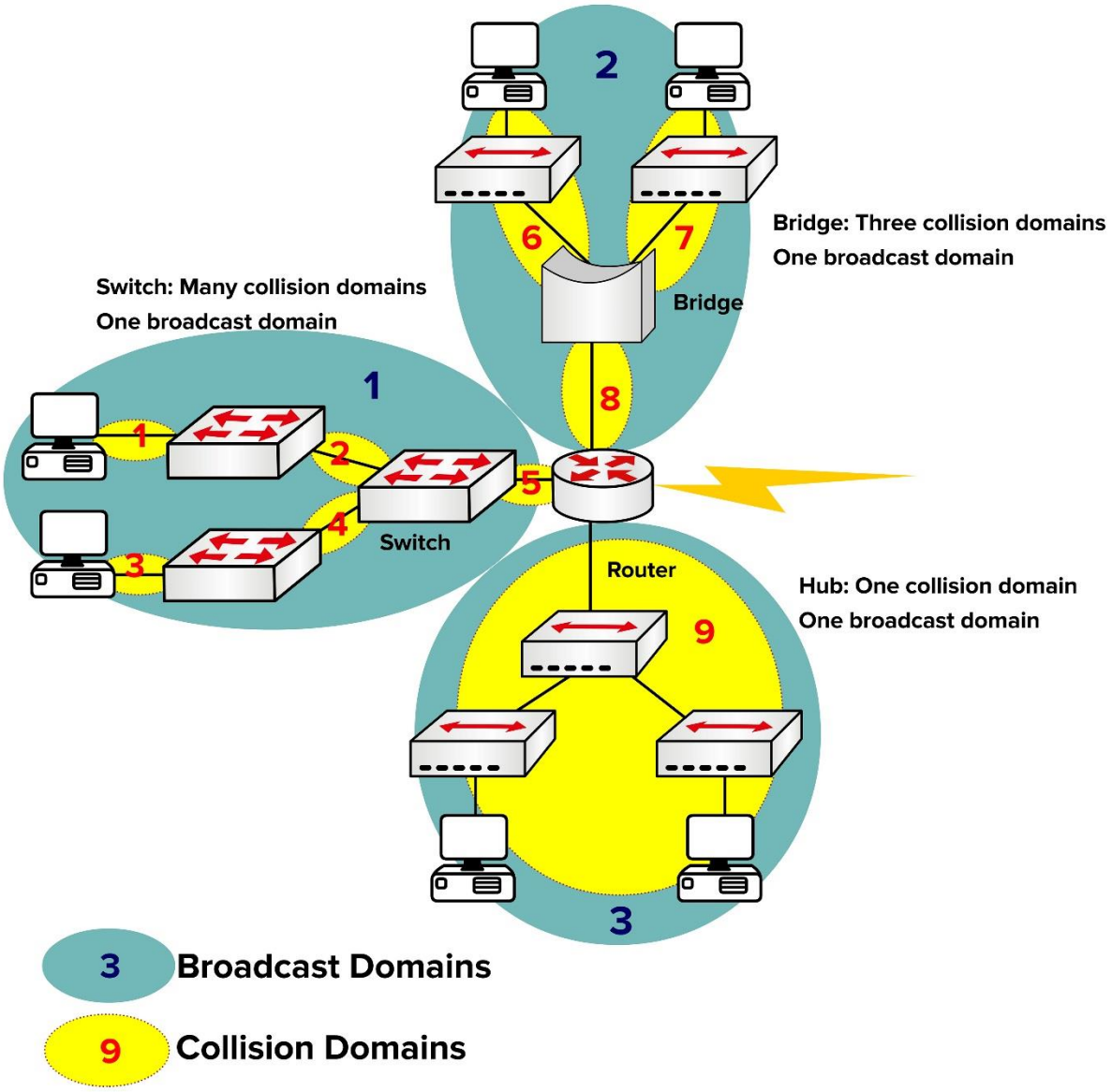
```
show vlan brief
show interface x
show interface x switchport
show interfaces trunk
```

Always ensure that you can ping across directly connected router interfaces BEFORE applying routing protocols and access lists. You have been warned.

2.0 Network Access

LAN Switching

Each port on a switch is a collision domain, meaning a damaged frame won't pass any further. Adding switches increases a number of collision domains on your network segment (which is a good thing). Hubs (and connected devices) are in one big collision domain, but router interfaces will not forward these frames.



A LAN switch has three primary functions:

1. Address Learning – maintains a table (CAM – Content Addressable Memory) of addresses and which port they can be reached on.
2. Forward/filter decision – forwards frames only out of the relevant port.
3. Loop avoidance - STP

Address Learning

Broadcast frames are forwarded out of all ports. Since Ethernet hosts can all transmit at the same time, this can lead to collisions, thus slowing down the network considerably.

When a switch is first powered up, it is not aware of the location of any of the hosts in the network. In a very short time, as hosts transmit data to other hosts, it learns and stores the source MAC address for each connected device. If an address is not currently in the switch's database, it will send a broadcast message out of each port (flooding), except the port that the request was received on. When the switch receives a reply, it will add the address and source port to its database. It can take only a matter of minutes to build this database. Cisco refers to this as the CAM (content-addressable memory) table.

The switch will store a table of MAC addresses for a limited amount of time. If no traffic is heard from that port for a predefined period of time, then the entry is purged from memory. This frees up memory space on the switch and prevents entries from becoming out of date and inaccurate. This sequence of actions is known as the MAC address aging time. On the Cisco 2960 model, this time period is 300 seconds by default, but it can be configured to be between 10 and 1,000,000 seconds.

```
Switch#show mac address-table aging-time
Global Aging Time:  300
Vlan    Aging Time
----    ---------
Switch#conf t
Switch(config)#mac-address-table aging-time 600
```

The switch can also be configured to never purge the addresses.

The command you would use to see the CAM table of a switch is show mac-address-table (note the dashes in the command).

Transmitting Frames Through a Switch

Store-and-Forward – switch copies the entire frame into its buffer and computes the CRC. The frame is discarded if there is an error. High latency.

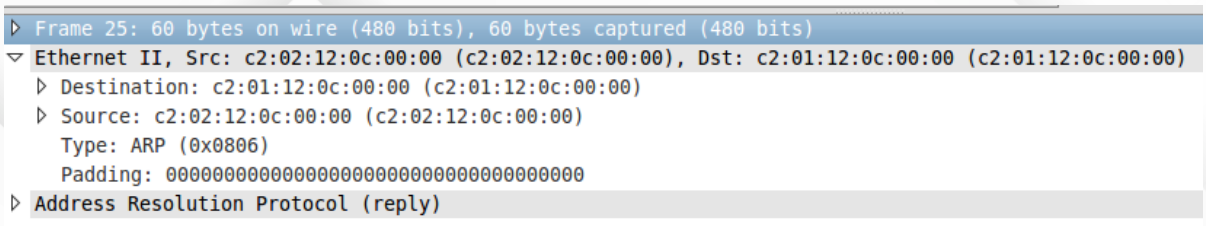
Cut-through – reads only the destination address (first 6 bytes after preamble), looks up address and forwards frame. Lower latency.

Fragment free – switch reads first 64 bytes before forwarding the frame. Collisions normally occur within the first 64 bytes.

Filtering and Forwarding Frames

Whenever a frame arrives at a switch port, the switch examines the destination address of the frame and then its database of MAC addresses. If the destination address is in the database, the frame will only be sent out of the interface the destination host is attached to. This process is known as frame filtering. If the address is not known, then the switch has no option but to flood the frame out of all ports other than the one on which it arrived.

Figure below shows a capture of a frame. The Destination field contains the address c201:120c:0000, and this is what the switch will look for in its MAC address table. If this address isn't present, it will broadcast for it.



Virtual LAN (VLAN)

A VLAN is a switched network that consists of logically segmented communities without regard for physical location. Each port on a switch can belong to a VLAN; by default, all ports are in VLAN 1. VLAN ports share broadcasts.

A router is needed to route traffic between VLANs because Layer 2 devices do not use IP addresses. VLANs reduce admin costs, tighten security and better control broadcasts.

VLAN trunk links are used to carry data from multiple VLANs. To differentiate one VLAN frame from another, all frames sent across a trunk link are specially tagged so that the destination switch knows which VLAN the frame belongs to. The switch used in the exam is a 2960, which only supports encapsulation type 802.1Q for trunk links.

The following summarizes some features of 802.1Q:

- Supports up to 4096 VLANs
- Uses an internal tagging mechanism, modifying the original frame
- Open standard protocol developed by the IEEE
- Does not tag frames on the native VLAN; however, all other frames are tagged

VLAN1 is the default VLAN for all ports and is not tagged. The untagged VLAN is referred to as the native VLAN.

Trunking

A trunk link is able to carry traffic from more than one VLAN. Although the link can be 10Mbps, the minimum recommended speed of the interface is 100Mbps. You can manually set the port to trunk (see below) or let the Dynamic Trunking Protocol (DTP) do this. DTP port settings can be dynamic auto or dynamic desirable and the default settings vary per switch model.

Settings which will result in a trunk forming are:

Desirable – Auto
Desirable - Desirable
Desirable – Trunk (manual)

You can see the default or current settings with the show interface X switchport command.

```
Switch#show int f0/2 switchport
Name: Fa0/2
Switchport: Enabled
Administrative Mode: dynamic auto
Operational Mode: down
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
```

Manually disable DTP on the interface with:

```
Switch1(config-if)#switchport nonegotiate
```

VLAN Trunking Protocol (VTP)

The VLAN Trunking Protocol (VTP) is a Cisco proprietary Layer 2 messaging protocol that manages the addition, deletion, and renaming of VLANs on switches in the same VTP domain. VTP allows VLAN information to propagate through the switched network, which reduces administration overhead in a switched network whilst enabling switches to exchange and maintain consistent VLAN information.

Switches are in server mode by default. In Server mode, the switch is authorized to create, modify, and delete VLAN information for the entire VTP domain. Any changes you make to a server are propagated throughout the whole domain. In Client mode, the switch will receive VTP information and apply any changes, but it does not allow adding, removing, or changing VLAN information on the switch. In Transparent mode, the switch will forward the VTP information received out of its trunk ports, but it will not apply the changes. Transparent mode is needed to configure the extended VLAN range (1006 to 4096).

VTP Pruning

Pruning prevents unnecessary VLAN information being sent over a trunk link if no interfaces are in the VLANs. Configured with the below command:

```
Switch(config)#vtp pruning
```

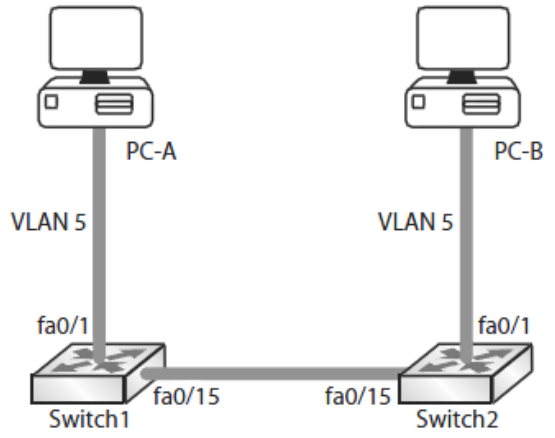
Common Switching Commands

This is not a sample configuration but a demonstration of commands you need to know.

```
Switch(config)#vlan 2  ← creates VLAN 2
Switch(config-vlan)#name SALES ← names VLAN
Switch(config)#interface fast 0/1
Switch(config-if)#switchport access vlan 2 ← puts interface into VLAN
2
Switch(config-if)#switchport mode trunk ← sets interface to trunk
Switch(config)#vtp mode transparent/client/server ← sets VTP mode
Switch(config)#vtp domain howtonetwork.com ← sets VTP domain name
Switch(config)#vtp password cisco
Switch(config)#ip default-gateway 192.168.1.1 ← switch default gateway
Switch#show vlan brief ← shows summary of VLAN info
Switch#show vtp status ← shows various VTP info including mode/version
Switch#show interfaces trunk ← shows trunk interfaces
Switch#show interfaces trunk ← shows trunk interfaces
Switch#show mac-address-table dynamic ← shows mac table (dynamic)
```

The switch must be in Transparent mode to add higher numbered VLANs (1006-4096), i.e. extended VLANs.

Sample configuration. Repeat commands on Switch1.



```
Switch1(config)#vlan 5
Switch1(config)#interface fast0/1
Switch1(config-if)#switchport mode access
Switch1(config-if)#switchport access vlan 5
Switch1(config-if)#interface fast0/15
Switch1(config-if)#switchport mode trunk
```

You can have both a voice VLAN and data VLAN on the same switch interface. Below is the configuration. CDP must be enabled on the switch or at least the interface in order for the voice VLAN to work.

```
SW1(config)#vlan10
SW1(config-vlan)#vlan20
SW1(config-vlan)#interface FastEthernet0/6
SW1(config-if)#switchport mode access
SW1(config-if)#switchport access vlan20
SW1(config-if)#switchport voice vlan10
```

Native VLAN

The IEEE defined the native VLAN to provide for connectivity to old 802.3 ports that did not understand VLAN tags. By default, VLAN1 is the native VLAN. You can change this with the command:

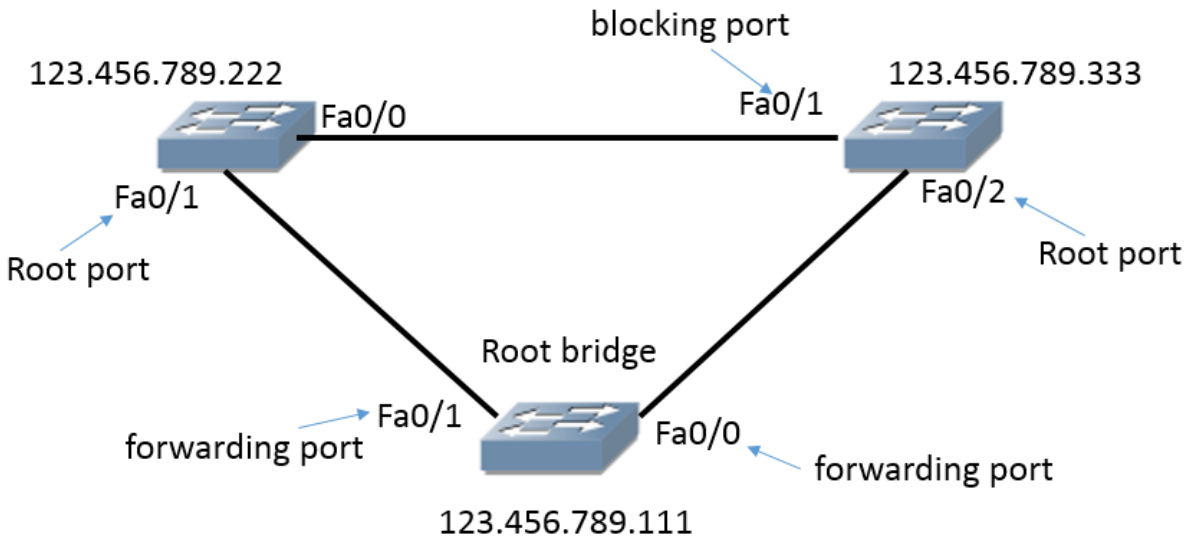
```
Switch(config-if) #switchport trunk native vlan 10
```

The native VLAN must match on either side of a trunk link.

LAN Switching

Spanning Tree Protocol (STP) IEEE 802.1d

STP is a link management protocol that provides path redundancy whilst preventing undesirable loops in the network. For communication to work correctly on an ethernet network, there can only be one path between two destinations. STP uses Bridge Protocol Data Units (BPDU) received by all switches to determine the spanning-tree topology. A port on a switch is either in forwarding or blocking state. Forwarding ports provide the lowest cost path to the root bridge, a port will remain in blocking state from start-up if spanning tree determines there is a better path.



You can force a particular switch to become a root bridge by manually configuring the priority (in increments of 4096):

```
Switch(config)#spanning-tree vlan 2010 priority 8192
```

Or

```
Switch(config)#spanning-tree vlan 2010 root primary
```

Port States in STP

Switch ports (the interfaces on the switch) running STP can be in one of five states. Please ensure that you understand all of these for the CCNA exam:

1. Blocking
2. Listening
3. Learning
4. Forwarding
5. Disabled

User data is only passed by a port in forwarding state

PortFast

PortFast is a feature that is typically enabled only for a port or interface that connects to a host. When the link comes up on this port, the switch skips the first stages of the STA and directly transitions to the Forwarding state.

```
Switch(config-if) #spanning-tree portfast
```

STP Security

As you have learned, PortFast disables STP on a switch port, but an important fact is that a PortFast switch port will keep listening for BDPUs. If someone adds a switch to a port that has been configured as PortFast, the consequences will be unpredictable and, in some cases, disastrous. To guard against this situation, Cisco provides the BPDU Guard, BPDU Filter, and Root Guard features

BPDU Guard

The BPDU Guard feature is used to protect the Spanning Tree domain from external influence. BPDU Guard is disabled by default but is recommended for all ports on which the Port Fast feature has been enabled. When a port that is configured with the BPDU Guard feature receives a BPDU, it immediately transitions to the err-disable state.

```
Switch(config-if) #spanning-tree bpduguard
```

BPDU Filter

When BPDU Filter is configured on a switch port that has been configured as PortFast, it will prevent the port from sending and receiving BPDUs on that port. This effectively disables STP on the port. This is unlike the behavior seen with BPDU Guard, where the port is put into an err-disabled mode. BPDU Filter can be enabled on the switch port using the spanning-tree bpdufilter enable command:

Loop Guard

Cisco Loop Guard is a Spanning Tree Protocol (STP) enhancement on Cisco switches that protects against Layer 2 forwarding loops by monitoring non-designated (blocking) ports for Bridge Protocol Data Units (BPDUs). If a blocking port stops

receiving expected BPDUs—often due to a unidirectional link failure—it transitions to a "loop-inconsistent" state to prevent accidental forwarding and loop formation, rather than defaulting to the listening/learning states. This feature is particularly useful in fiber optic environments prone to one-way link issues and is configured globally or per-port on Cisco IOS devices.

Root Guard

Root Guard is configured per port and, as with BPDU Guard, it monitors for incoming BPDUs. Root Guard is designed to prevent the port from becoming a root port. If a superior BPDU is received on the port, the port is placed into a root-inconsistent state, preventing it from forwarding or receiving frames until the superior BPDUs cease

Rapid Spanning Tree Protocol (RSTP) IEEE 802.1w

The spanning tree takes up to 50 seconds to converge to a stable network whereas RSTP takes 2 seconds. RSTP port roles are root port, designated port, backup port, alternate port and disabled. Most implementations of RSTP use PVST+, Per VLAN Spanning Tree+, here multiple instances of Spanning Tree are running so the load on the CPU is higher but we can load share over the links.

To enable RSTP for each VLAN in our switched network we use the following command:

```
Switch(config)#spanning-tree mode rapid-pvst
```

Extended VLANs

Extended VLANs are not stored in the vlan.dat file (i.e. the VLAN database) and are not advertised via VTP. If you want to configure them, you must put the switch into VTP transparent mode (or disable VTP completely). Extended VLANs are numbered from 1006 to 4094, inclusive. You won't actually see them listed when you issue the show vlan brief command.

EtherChannels

An EtherChannel is comprised of physical, individual FastEthernet, GigabitEthernet, or Ten-GigabitEthernet (10Gbps) links that are bundled together into a single logical link.

Each EtherChannel can consist of up to eight ports. Physical links in an EtherChannel must share similar characteristics, such as being defined in the same VLAN or have the same speed and duplex settings, for example.

There are two link aggregation protocol options that can be used to automate the creation of an EtherChannel group: Port Aggregation Protocol (PAgP) and Link Aggregation Control Protocol (LACP). PAgP is a Cisco proprietary protocol, while LACP is part of the IEEE 802.3ad specification for creating a logical link from multiple physical links.

Port Aggregation Protocol (PAgP) is a Cisco proprietary link aggregation protocol that enables the automatic creation of EtherChannels. By default, PAgP packets are sent between EtherChannel-capable ports in order to negotiate the forming of an EtherChannel. These packets are sent to the destination Multicast MAC address 01-00-0C-CC-CC-CC, which is also the same Multicast address that is used by CDP, UDLD, VTP, and DTP.

Switch EtherChannels using PAgP may be configured to operate in one of two modes: auto or desirable. At least one side must be desirable in order for the link to come up. LACP packets are sent to the IEEE 802.3 Slow Protocols Multicast group address 01-80-C2-00-00-02. LACP supports two modes: active and passive. At least one side must be active.




```
Switch1(config)#int range f0/1-2
Switch1(config-if-range)#switchport mode access
Switch1(config-if-range)#channel-group ?
  <1-6>  Channel group number
Switch1(config-if-range)#channel-group 1 mode ?
  active     Enable LACP unconditionally
  auto       Enable PAgP only if a PAgP device is detected
  desirable  Enable PAgP unconditionally
  on         Enable Etherchannel only
  passive    Enable LACP only if a LACP device is detected
Switch1(config-if-range)#channel-group 1 mode desirable
```

InterVLAN Routing SVI

SVIs are logical interfaces that represent a VLAN. Although the SVI represents a VLAN, it is not automatically configured when a Layer 2 VLAN is configured on the switch; it must be manually configured by the administrator using the interface vlan [number] global configuration command. The Layer 3 configuration parameters, such as IP addressing, are then configured on the SVI in the same manner as they would be on a physical interface.

The following output illustrates the configuration of SVIs to allow inter-VLAN routing on a single switch. This output references the VLANs used in the previous configuration outputs in this section:

```
Switch(config)#vlan 10
Switch(config-vlan)#name Example-VLAN-10
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name Example-VLAN-20
Switch(config-vlan)#exit
Switch(config)#interface range FastEthernet0/1 - 2
Switch(config-if-range)#switchport
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#interface range FastEthernet0/3 - 4
Switch(config-if-range)#switchport
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#interface vlan 10
Switch(config-if)#description "SVI for VLAN 10"
Switch(config-if)#ip address 10.10.10.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#description 'SVI for VLAN 20'
Switch(config-if)#ip address 10.20.20.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
```

3.0 IP Connectivity

IP Routing

Routers must have some means of learning networks that are not directly connected. The show ip route command provides information regarding the routing tables. It lists the networks the router can reach and information about the way they can be reached, including:

- Network
- Routing Protocol
- Next hop
- Outgoing interface

Some protocols will use a topology table containing alternative paths to the same network which could be used should the main path fail.

```
R1#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
       area, N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
       type 2, E1 - OSPF external type 1, E2 - OSPF external type 2,
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
       inter area, * - candidate default, U - per-user static route,
       o - ODR, P - periodic downloaded static route
Gateway of last resort is not set
R       80.1.1.0/24 [120/1] via 10.1.1.2, 00:00:04, Ethernet0/0.1
D       80.0.0.0/8 [90/281600] via 10.1.1.2, 00:02:02, Ethernet0/0.1
O E2    80.1.0.0/16 [110/20] via 10.1.1.2, 00:00:14, Ethernet0/0.1
```

Prefix Matching

Routers use the longest prefix match rule when determining which of the routes placed into the routing table should be used to forward traffic to a destination network or node. Longer or more specific routing table entries are preferred over less specific entries, such as summary addresses, when determining which entry to use to route traffic.

The longest prefix or the most specific route will be used to route traffic to the destination network or node, regardless of the administrative distance of the route source, or even the routing protocol metric assigned to the prefix if multiple overlapping prefixes are learned via the same routing protocol. Table 13-1 below illustrates the order of route selection on a router sending packets to the address 1.1.1.1. This order is based on the longest prefix match lookup.

Table 13-1: Matching the longest prefix

Routing Table Entry	Order Used
1.1.1.1/32	1st
1.1.1.0/24	2nd
1.1.0.0/16	3rd
1.0.0.0/8	4th
0.0.0.0/0	5th

NOTE: Although the default route is listed last in the route selection order in Table 13-1, keep in mind that a default route is not always present in the routing table. If that is the case and no other entries to address 1.1.1.1 exist, packets to that destination are simply discarded by the router. In most cases, the router will send the source host an ICMP message indicating that the destination is unreachable.

Static routing

```
Router(config)#ip route {destination network}{mask}{next hop address}
e.g ip route 172.16.5.2 255.255.255.0 172.16.12.8
```

or exit interface:

```
ip route 172.16.5.2 255.255.255.0 s0/0
```

The ip default-gateway command should only be used when IP routing is disabled on the Cisco router. It is commonly used on a Layer 2 switch. You can use the ip default-network command when IP routing is enabled on the Cisco router. When you configure the ip default-network command, the router considers routes to that network for installation as the gateway of last resort on the router.

```
R1(config)#ip default-network 198.10.1.0
```

If you want the router to use a route if the routing protocol fails, you can install a floating static route and add an AD to the end, which is higher than that of the routing protocol.

```
Router(config)#ip route 192.168.1.0 255.255.255.0 172.16.5.3 115
```

The below default route will forward packets to unknown IPv6 destinations via Serial0/1 using the Link-Local address of the next-hop router as the IPv6 next-hop address. For broadcast interfaces (such as Ethernet), you need to add an exit interface.

```
R1(config)#ipv6 route ::/0 Serial0/1 FE80::3333
```

Distance Vector

Distance Vector protocols understand the direction and distance to any given network connections. Algorithms calculate the cost to reach the connection and pass this information to every neighbor router. Examples are RIP and IGRP. Problems with distance vector protocols include routing loops and counting to infinity. To overcome these problems, the following can be implemented:

Defining a maximum number of hops, 15 for RIP and 255 for IGRP.

Split Horizon – if the router learns a route on an interface, it does not advertise it out of the same interface.

Hold Down Timers – ignores new routing updates until a determined time has passed.

Triggered Updates – instead of routing updates being sent at the default intervals, a triggered update is sent every time to indicate a change in the routing table.

Link State

These have a picture of the entire network from link state advertisements (LSA) and link state packets (LSP). Once these have all been passed, only changes to the network are sent out reducing network traffic.

Link state protocols do require a lot of CPU time and bandwidth when LSAs are flooded out. Examples are OSPF and ISIS.

Routers use administrative distances to determine how believable the route learned is depending upon the protocol it learns the router from.

Source	Default Distance
Directly Connected Interface	0
Static hop to next router	1
EIGRP Summary	5
External BGP	20
EIGRP (Internal)	90
OSPF	110
IS-IS	115
RIP	120
Exterior Gateway Protocol (EGP)	140
External EIGRP	170
Internal BGP	200
Unknown	255

An administrative distance of 0 is most preferred, so a router running RIP and OSPF will prefer the OSPF routes most and install these in the routing table. Cisco routers use the longest prefix match rule when determining which of the routes placed into the routing table should be used to forward traffic to a destination network or node so the longest match from RIP will be preferred over a shorter match from OSPF (for example).

Routing protocols maintain a table of networks and which interface they can be reached by. Examples: RIP, OSPF. Routed protocols are used to transport traffic from source to destination. Examples: IP, IPX, Appletalk.

BGP is an exterior gateway protocol. It is used to connect autonomous systems together.

When a packet traverses the network from device to device (hop to hop), the IP address remains constant, the hardware (MAC) address changes.

You may wish to stop routing updates from being sent on a certain interface. The command for accomplishing this is usually passive-interface [interface number]. You will still receive routing updates on this interface.

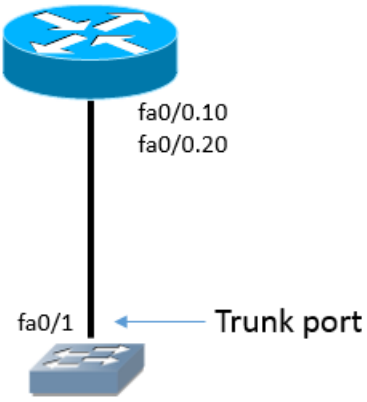
Inter-VLAN Routing

Using a Layer 3 switch with SVIs or Router with a VLAN trunk connecting switch.

- Create sub-interface for each VLAN that is required to be routed.
- Use 802.1Q and associate VLAN with sub-interface.
- Configure IP address for each.
- Also called router-on-a-stick

Example:

```
R1(config)#interface fast 0/0.10
     encapsulation dot1q 10
     ip address 10.1.1.1 255.255.255.0
     !
     interface fast 0/0.20
     encapsulation dot1q 20
     ip address 20.1.1.1 255.255.255.0
     !
     interface fast 0/0.30
     encapsulation dot1q 30
     ip address 30.1.1.1 255.255.255.0
```



The switch will need to have VLANs created, ports put into the relevant VLANs and a trunk link configured to the switch and the router interface. Add a default gateway to the router which will be VLAN 30 in the above configuration.

OSPF

- Uses IP protocol 89
- Classless
- Uses Dijkstra's shortest path first algorithm (SPF)
- Router ID is the highest IP address but loopback address used if present
- Backbone area is area 0
- All non-backbone areas must connect directly to area 0
- Areas can be numbered from 0 to 65535
- Multicasts on 224.0.0.5

OSPF basic terms:

- LSA – OSPF uses Link State Advertisement (LSA) to organize the topology information; it is a data structure with specific information about the network.
- LSDB – Link State Database (LSDB) is the collection of LSAs known to the router.
- ABR – In multiple area OSPF designs, some routers sit at the border of multiple OSPF area so they are known as Area Border router (ABR).
- Router ID - Every router in the OSPF network is identified by router ID which can be manually assigned or automatically based on highest IP address of physical/loopback interface on the router.

Interface	Cost (108/Bandwidth)
ATM, Fast Ethernet, Gigabit Ethernet, FDDI (> 100 Mbps)	1
HSSI (45Mbps)	2
16 Mbps Token Ring	6
10 Mbps Ethernet	10
4 Mbps Token Ring	25
T1 (1.544 Mbps)*	64
DS-0 (64k)*	1562
56k	1785

OSPF Network types:

- BMA – Broadcast Multi-access
- Point to Point
- Point to Multipoint

OSPF maintains three tables:

i. Neighbor table
ii. Database table
iii. Routing information Table

DR: Designated Router is selected on the basis of higher priority number which is 1 by default or on the basis of highest router ID.

BDR: Backup Designated Router is selected on the basis of second highest priority or second highest router ID.

OSPF LSA 7 types:

LSA Type 1 - Router LSA
LSA Type 2 - Network LSA
LSA Type 3 - Summary LSA
LSA Type 4 - ASBR summary LSA
LSA Type 5 - External LSA
LSA Type 6 - Multicast LSA
LSA Type 7 - NSSA LSA

```
Router(config)#router ospf 20
Router(config-router)#network 172.16.0.0 0.0.255.255 area 0
```

Note that the wildcard must match the subnet you are advertising so if you have network 192.168.4.0 you need to advertise:

```
network 192.168.4.0 0.0.0.3 area 0
```

NOT

```
network 192.168.1.0 0.0.0.3 area 0
```

First Hop Redundancy Protocol

First hop routing protocol provides protection from default gateway failover by providing backup routers. In the event of a failover, the backup router takes the place of the active router; it provides redundancy where required and also load balancing.

By default, when Hot Standby Router Protocol is enabled in Cisco IOS software, version 1 is enabled. HSRP version 1 restricts the number of configurable HSRP groups to 255. HSRP version 1 routers communicate by sending messages to Multicast group address 224.0.0.2 using UDP port 1985.

HSRP version 2 uses the new Multicast address 224.0.0.102 to send Hello packets instead of the Multicast address of 224.0.0.2, which is used by version 1. The UDP port number, however, remains the same.

In HSRP version 1, the Layer 2 address that is used by the virtual IP address will be a virtual MAC address composed of 0000.0C07.ACxx, where “xx” is the HSRP group number in hexadecimal value and is based on the respective interface. HSRP version 2, however, uses a new MAC address range of 0000.0C9F.F000 to 0000.0C9F.FFFF for the virtual gateway IP address.

HSRP primary gateway election can be influenced by adjusting the default HSRP priority of 100 to any value between 1 and 255. The router with the highest priority will be elected as the primary gateway for the HSRP group.

In order for a gateway with a higher priority to assume active gateway functionality when a primary gateway is already present for an HSRP group, the router must be configured for pre-emption.

```
VTP-Server-1(config)#interface vlan172
VTP-Server-1(config-if) #ip address 172.16.31.1 255.255.255.0
VTP-Server-1(config-if) #standby 1 ip 172.16.31.254
VTP-Server-1(config-if) #standby 1 priority 105
VTP-Server-1(config-if) #standby 1 preempt
VTP-Server-1(config-if) #exit
```

VRRP operates in a similar manner to HSRP; however, unlike HSRP, VRRP is an open standard that is defined in RFC 2338, which was made obsolete by RFC 3768. VRRP sends advertisements to the Multicast destination address 224.0.0.18 (VRRP), using IP protocol number 112. At the Data Link layer, advertisements are sent from the master virtual router MAC address 00-00-5e-00-01xx, where 'xx' represents the two-digit Hexadecimal group number.

When GLBP is enabled, the GLBP group members elect one gateway to be the active virtual gateway (AVG) for that group. The AVG is the gateway that has the highest priority value. In the event that the priority values are equal, the AVG with the highest IP address in the group will be elected as the gateway.

4.0 IP Services

NAT

Network Address Translation will convert and address from the inside of your network to another address on the outside of your network and vice versa. It is most commonly used to convert a non-routable address to a routable address.

For all configurations, you must specify which interfaces are internal for NAT and which are external:

```
Router(config-if) #ip nat inside/outside
```

Static NAT – maps one address to one address such as 192.168.1.1 to 200.1.1.1. You would do this for any devices which need a permanent IP address, such as a web server.

```
Router(config)#ip nat inside source static 192.168.1.1 200.1.1.1
```

Dynamic NAT – maps a number of internal addresses to a pool of external addresses. The below configuration creates a pool of 10 addresses with a mask (prefix length) of 255.255.255.0 and the name 'ad_team.' The hosts to be NATted are on the 192.168.1.0 network. The access list (source list) tells the router which addresses to NAT.

```
Router(config)#ip nat pool ad_team 10.0.0.1 10.0.0.10 prefix-length 24
Router(config)#ip nat inside source list 1 pool ad_team out
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
```

Overload NAT – (or PAT) maps private internal addresses to one or more external addresses using port numbers. The below configuration creates a pool of ten addresses (it could be more) and the command overload tells the router to use port address translation.

```
Router(config)#ip nat pool ad_team 10.0.0.1 10.0.0.10 prefix-length 24
Router(config)#ip nat inside source list 1 pool ad_team out overload
Router(config)#access-list 1 permit 192.168.1.0 0.0.0.255
Router#show flash:
```

QoS

QoS lets you manage scarce bandwidth so you get the most out of what you have and provide predictable performance. The default behavior is verified using the show mls qos command as shown in the following output:

```
Switch#show mls qos
QoS is disabled
Switch#conf t
Switch(config)#mls qos
Switch(config)#end
Switch#show mls qos
QoS is enabled
QoS ip packet dscp rewrite is enabled
```

QoS provides three broad delivery models to consider when designing and implementing a network infrastructure.

1. Best-effort delivery (default)
2. Integrated services
3. Differentiated services

Integrated services (IntServ) were defined in RFC 1633, and is a set of extensions of the best-effort model. It applies a policy per flow of traffic and explicitly manages network resources to deliver end-to-end QoS. Differentiated services (DiffServ) defines the concept of a service class. Each device handles the packet individually (on a per-hop basis). Layer 2 DiffServ uses Class of Service (CoS) bits encapsulated into an 802.1Q frame (in the VLAN field).

Ingress QoS mechanisms are applied to frames and packets received by the switch in the inbound direction. The following Catalyst Switch ingress QoS mechanisms will be described in the following sections:

- Traffic classification
- Traffic policing
- Marking
- Congestion management and avoidance

Classification is used to differentiate one stream of traffic from another so that different service levels can be applied to different streams of traffic. Frames can be classified based on the incoming CoS or DSCP values or even based on access control list (ACL) configuration.

Policing is a process that is used to limit traffic to a prescribed rate. Policing is used to compare the ingress/egress traffic rate to a configured policer. The policer is configured with a rate and a burst. Marking involves setting QoS bits inside the Layer 2 or Layer 3 headers, which allows the other internetwork devices to classify based on the marked values. Marking is typically used in conjunction with traffic policing.

Congestion management and avoidance are comprised of the following three elements:

- Queuing
- Dropping
- Scheduling

Queuing is used to place packets into different software queues based on the QoS labels. After the traffic is classified and marked with QoS labels, it is assigned and placed into different queues based on those QoS labels

SNMP

SNMP is an Application Layer (Layer 7) protocol, using UDP ports 161 and 162, that facilitates the exchange of management information between network devices. A SNMP-managed network consists of a management system, agents, and managed devices. The management system executes monitoring applications and controls managed devices.

An SNMP agent resides on each managed device and translates local management information data, such as performance information or event and error information caught in software traps, into a readable form for the management system. SNMP agents use get-requests that transport data to the network management software. SNMP agents capture data from Management Information Bases (MIBs), which are device parameter and network data repositories, or from error or change traps.

A managed element, such as a router, a switch, a computer, or a firewall, is accessed via the SNMP agent. Managed devices collect and store management information, making it available through SNMP to other management systems having the same protocol compatibility.

When using SNMP, managed devices are monitored and controlled using three common SNMP commands: read, write, and trap. The read command is used by an NMS to monitor managed devices. This is performed by the NMS, examining different variables that are maintained by managed devices. The write command is used by an NMS to control managed devices. Using this command, the NMS can change the values of variables stored within managed devices. Finally, the SNMP trap command is used by managed devices to report events to the NMS.

The three versions of SNMP are versions 1, 2, and 3. Version 1, or SNMPv1, is the initial implementation of the SNMP protocol. SNMPv1 operates over protocols such as User Datagram Protocol (UDP), Internet Protocol (IP), and the OSI Connectionless Network Service (CLNS).

SNMPv2 revises SNMPv1 and includes improvements in the areas of performance, security, confidentiality, and manager-to-manager communications. SNMPv2 also defines two new operations: GetBulk and Inform. The GetBulk operation is used to retrieve large blocks of data efficiently.

SNMPv3 provides the following three additional security services that are not available in previous versions of SNMP: message integrity, authentication, and encryption. SNMPv3 uses message integrity to ensure that a packet has not been tampered with in-transit. SNMPv3 also utilizes authentication, which is used to determine whether the message is from a valid source. Finally, SNMPv3 provides encryption, which is used to scramble the contents of a packet to prevent it from being seen by unauthorized sources.

```
R2#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#snmp-server community unsafe RO
R2(config)#snmp-server community safe RW
R2(config)#snmp-server host 1.1.1.1 traps readonlypassword rtr syslog
```

5.0 Security Fundamentals

Network Security

The principle of least privilege in data security states that each user should have the rights necessary to do all they are required to do, and no more. Users must be made aware of policies that directly impact them. An acceptable use policy will not impact the behavior of anyone who is not aware of it. Users should undergo frequent security policy training and understand related procedures.

A good practice is to have frequent (usually annual) security-related awareness training and to have users sign a user agreement. This leaves no doubt as to the expectations of user conduct. Keeping users aware of and compliant with security policies minimizes the organizational risk posed by users.

Switch and Router Security

Passwords

The command `service password-encryption` encrypts all passwords with a weak (level 7) algorithm.

Enable: used to get from user exec to privileged exec. Not encrypted.

```
Router(config)#enable password {password}
```

Enable Secret: Encrypts password (only use enable or enable secret not both). Level 5 encryption.

```
Router(config)#enable secret {password}
```

VTY: needed if telnet access is required.

```
Router(config)#line vty 0 4
Router(config-line) #password cisco
Router(config-line) #login
```

If you want to permit SSH into the router or switch telnet lines, then you need to add the command transport input ssh to the vty lines. In order to generate an RSA key for SSH, we need to configure the hostname and a DNS domain name on the router (a username and password is also required). Therefore, in fact, both answer C and answer D are correct.

Auxiliary: allows modem access to the aux port.

```
Router(config)#line aux 0
Router(config-line) #password cisco
Router(config-line) #login
```

Console: used to allow console access

```
Router(config)#line console 0
Router(config-line) #password cisco
Router(config-line) #login
```

Protect Switch Ports

```
Switch1(config)#int fast 0/1
Switch1(config-if) #switchport mode access
Switch1(config-if) #switchport port-security
Switch1(config-if) #switchport port-security?
  mac-address Secure mac address
  maximum     Max secure addresses
  violation   Security violation mode
  <cr>
Switch1(config-if) #sw port-security mac-address sticky
```

The port must be statically set to either trunk or access, you can't apply switch security to a dynamic port. If you add the sticky command to the configuration, the switch will store the learned mac addresses in the running configuration.

Violation Action is the action the switch will take should a violation take place. By default, it's set to shutdown.

```
Switch1(config-if)#switchport port-security violation ?
  protect   Security violation protect mode
  restrict  Security violation restrict mode (sends SNMP trap)
  shutdown  Security violation shutdown mode (sends SNMP trap)
```

Mode	Port Action	Traffic	Syslog	Violation Counter
Protect	Protected	Unknown MACs discarded	No	No
Shutdown	Errdisabled	Disabled	Yes and SNMP trap	Incremented
Restrict	Open	# of excess MAC traffic denied	Yes and SNMP trap	Incremented

A series of events can cause Cisco switches to put their ports into a special disabled mode called err-disabled. This basically means that a particular port has been disabled (shut down) due to an error. Fix by bouncing the interface (shutdown and then no shutdown).

VTP Password:

```
Switch1(config)#vtp password cisco
```

Restrict VLANS Passing on Ports:

```
Switch1(config-if)#switchport trunk allowed vlan 7-12
```

IPv4 Access Lists

Access lists are a set of conditions that permit or deny access to or through a router's interface.

Range	Usage
1-99	IP Standard
1300-1999	IP Standard (Expanded Range)
100-199	IP Extended
2000-2699	IP Extended (Expanded Range)

Standard Access Lists

Standard IP access lists check only the source address of the packet and permits or denies the entire TCP/IP suite. You cannot choose a particular port or application to block. Cisco recommends that they are placed as close to the destination as possible.

```
Router(config)#access-list{number 1-99}{permit/deny}{source address}
access-list 10 permit 172.16.5.2 ← address can be a host or network
```

Extended Access Lists

These allow for a lot more granularity when filtering IP traffic. They can filter packets based on source or destination, a particular IP protocol and port number. Cisco recommends that they are placed as close to the source as possible.

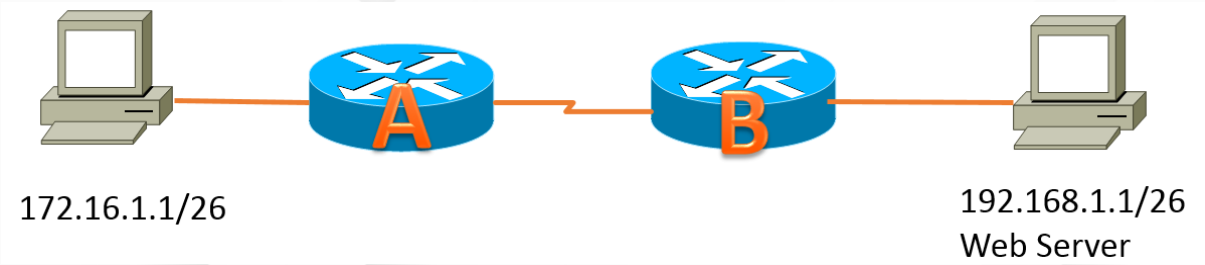
```
Router(config)#access-list {number 100-99} {permit/deny}{protocol}
{source}{destination}{port}

access-list 112 permit tcp host 172.16.5.2 host 172.16.10.2 eq www
```

Named Access Lists

```
Router(config)#ip access-list {standard/extended} name
Router(config)#ip access-list extended no_ftp
```

Sample Configs



EXTENDED ACL Router A

```
interface serial 0/0
access-group 100 in
!
access-list 100 permit tcp host 192.168.1.1 host 172.16.1.1 eq 80
```

STANDARD ACL Router A

```
interface serial 0/0
access-group 1 in
!
access-list 1 permit 192.168.1.1
```

NAMED ACL Router A

```
interface serial 0/0
ip access-list blockweb in
!
ip access-list extended blockweb
permit tcp host 192.168.1.1 host 172.16.1.1 eq 80
```

Access lists applied to inbound interfaces save the router having to process the packet, denied packets will be dropped at the interface. Outbound access lists will be processed by the router and then dropped at the outbound interface if they match the access list.

Access lists can be applied to multiple interfaces but there can only be one access list per protocol per direction per interface.

Use the term '`access-class`' if applying to console/aux/vty lines.

```
show ip access-lists
show access-list 1
```

Packets are processed by the access list and then routed.

Wildcard Masks

Wildcard masks tell the router which parts of the address to look at and which to disregard.

```
access-list 12 permit 172.16.5.0 0.0.0.255
```

This would permit any host on network 172.16.5.x. In order to work out a wildcard mask, simply write out the mask in full and then take that number away from 255.

Number	255	255	255	255
- Mask	255	255	192	0
Equals	0	0	63	255

Access lists are applied to interfaces:

```
Router(config)#access-list 1 permit 172.16.5.2
Router(config)#interface e0
Router(config-if) #ip access-group 1 in
```

Syslog

A syslog daemon or server is an entity that listens to the syslog messages that are sent to it. You cannot configure a syslog daemon to ask a specific device to send it syslog messages.

Syslog uses User Datagram Protocol (UDP) as the underlying transport mechanism, so the data packets are unsequenced and unacknowledged.

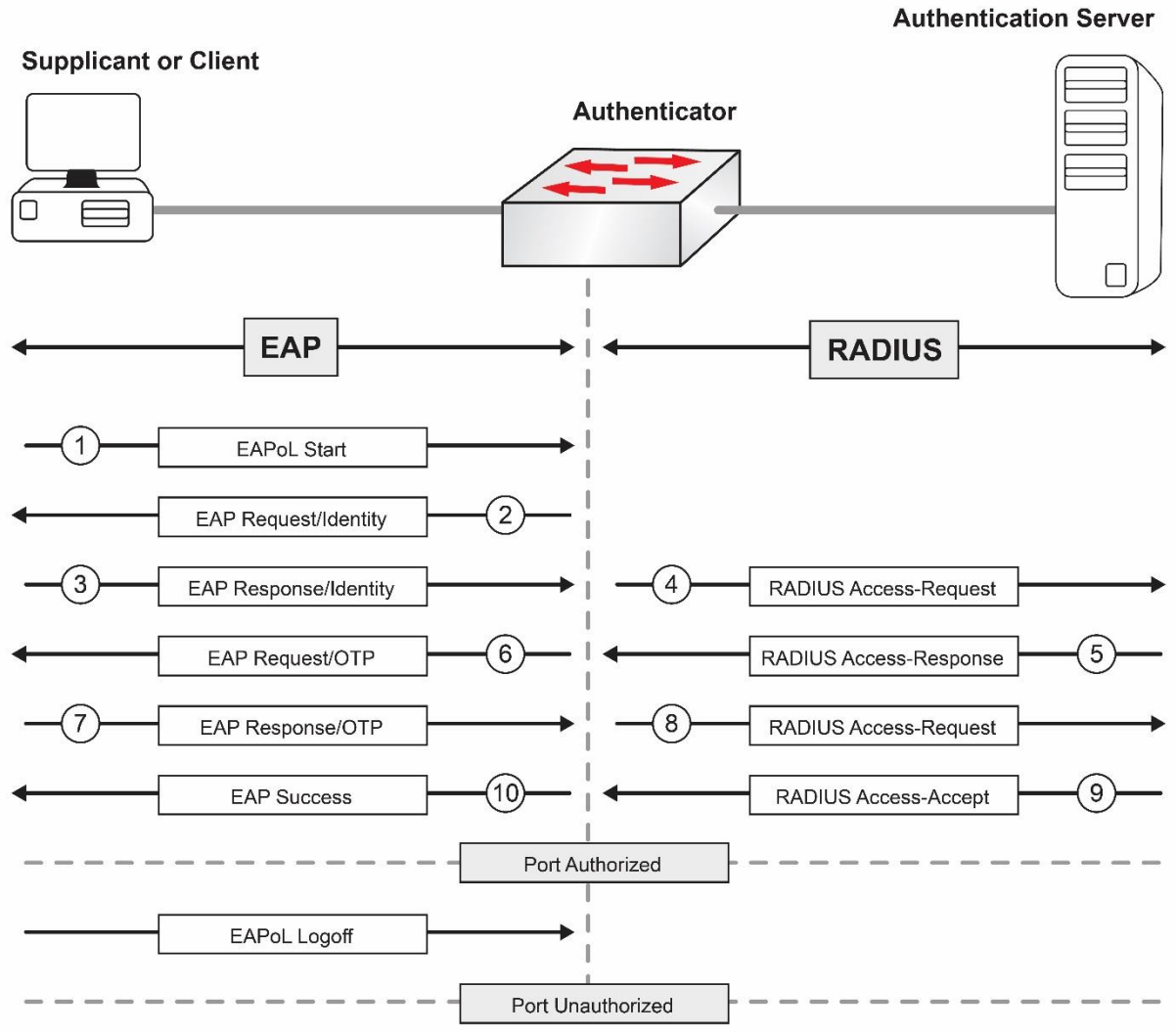
The Syslog client sends syslog messages to the syslog server using UDP as the Transport Layer protocol, specifying a destination port of 514. These messages cannot exceed 1024 bytes in size; however, there is no minimum length. All Syslog messages contain three distinct parts: the priority, the header, and the message.

Level	Level Name	Syslog Definition	Description
0	Emergencies	LOG_EMERG	This level is used for the most severe error conditions, which render the system unusable.
1	Alerts	LOG_ALERT	This level is used to indicate conditions that need immediate attention from administrators.
2	Critical	LOG_CRIT	This level is used to indicate critical conditions, which are less critical than Alerts but still require administrator intervention.
3	Errors	LOG_ERR	This level is used to indicate errors within the system; however, these errors do not render the system unusable.
4	Warnings	LOG_WARNING	This level is used to indicate warning conditions about system operations that did not complete successfully.
5	Notifications	LOG_NOTICE	This level is used to indicate state changes within the system (e.g. a routing protocol adjacency transitioning to a down state).
6	Informational	LOG_INFO	This level is used to indicate informational messages about the normal operation of the system.
7	Debugging	LOG_DEBUG	This level is used to indicate real-time (debugging) information that is typically used for troubleshooting purposes.

Access Layer Threats

802.1X provides Port-based Network Access Control (PNAC) and authenticates users or devices before they have access to the LAN or Wireless LAN. It provides port-level policy enforcement and then places users into a predetermined VLAN granting certain network rights.

802.1X employs the concepts of supplicant, authenticator, and the authentication server. Once implemented, the LAN switch acts as the authenticator, as shown in Figure 40 below. In order for a client (supplicant) to gain access to the network, he or she must be authenticated by the connected switch (authenticator), which in turn contacts the authentication server for authorization. For this to work, the end-user device must be using 802.1X client software, which is included with most major operating systems, such as Windows and Ubuntu.



DHCP snooping provides network protection from rogue DHCP servers by creating a logical firewall between untrusted hosts and DHCP servers. When DHCP snooping is enabled, the switch builds and maintains a DHCP snooping table (which is also

referred to as the DHCP binding table), and it is used to prevent and filter untrusted messages from the network.

DHCP snooping uses the concept of trusted and untrusted interfaces. This means that incoming packets received on untrusted ports are dropped if the source MAC address of those packets does not match the MAC address in the binding table.

```
Switch(config)#ip dhcp snooping
Switch(config)#ip dhcp snooping vlan100
Switch(config)#int gi2/24
Switch(config-if) #description "Connected to Legitimate DHCP Server"
Switch(config-if) #ip dhcp snooping trust
Switch#show ip dhcp snooping
Switch DHCP snooping is enabled.
DHCP Snooping is configured on the following VLANs:
100
Insertion of option 82 information is enabled.
Interface            Trusted        Rate limit (pps)
---------            -------        ----------------
GigabitEthernet2/24 yes             none
```

Dynamic ARP Inspection is a Catalyst switch security feature that validates ARP incoming (not outgoing) packets in a network. DAI determines the validity of packets by performing an IP-to-MAC address binding inspection. Once this validity has been confirmed, packets are then forwarded to their destination; however, DAI will drop all packets with invalid IP-to-MAC address bindings that fail the inspection validation process. DAI ensures that only valid ARP requests and responses are relayed and so protects the network from certain man-in-the-middle attacks.

AAA

Authentication is used to validate identity (i.e. who the user is); authorization is used to determine what that particular user can do (i.e. the services available to the user); and accounting is used for audit trails (i.e. what that user did).

AAA uses standard authentication methods, which include Remote Authentication Dial-In User Service (RADIUS), Terminal Access Controller Access Control System Plus (TACACS+), and Kerberos. Once the security server has received the credentials, it will respond with a pass (accept) or fail (deny) message. Authentication also offers

additional services, such as challenge and response, messaging support, and even encryption, depending on the security protocol implemented.

In order for AAA to work, the Network Access Server (NAS), which is any device—a router, switch, or firewall—must be able to access security information for a specific user before providing AAA services. This information may be stored locally (i.e. on the NAS itself) or remotely (i.e. on a RADUIS, TACACS+, or Kerberos server).

RADIUS uses UDP as the Transport Layer protocol for communication between the client and the server, using UDP port 1812 for authentication and authorization and UDP port 1813 for accounting. Unlike RADIUS, which is an open-standard protocol, TACACS+ is a Cisco-proprietary protocol that is used in the AAA framework to provide centralized authentication of users who are attempting to gain access to network resources.

There are several notable differences between TACACS+ and RADIUS. One of the most notable differences is that TACACS+ uses TCP as a Transport Layer protocol, using TCP port 49. You would globally enable AAA services using the aaa new-model command in global configuration mode.

6.0 Automation and Programmability

Software Defined Network

SDN aims to disassociate the decisions about traffic flow from the systems that forward the network traffic. Network managers (using SDN) are now able to manage and control network behavior dynamically using open interfaces and application programming interfaces (APIs). In a nutshell, its aim is to move configuration and control of the network from hardware to software control.

Southbound APIs relay information to routers and switches. OpenFlow was the first to market and was the originator of the southbound API. The Southbound interface allows the controller and the network device to communicate. There are different types of controllers, each with different capabilities. Here are three examples that you should be aware of:

- OpenFlow (from the ONF; https://www.opennetworking.org)
- OpFlex (from Cisco; used with ACI)
- CLI (Telnet/SSH) and SNMP (from Cisco; used with APIC-EM)

Northbound APIs are used by SDN to communicate with applications and business logic. The northbound interface opens the controller, allowing data and functions access by programs.

Network devices are equipped with Data Plane, Control Plane, and Management Plane.

The Data Plane provides the ability of network devices to forward messages. The process of receiving, processing, and forwarding data is part of the Data Plane. To let the Data Plane know where to send, a device uses the information obtained from the control plane. An example of this is the process of switching data from one port of a switch to another. It uses the information in the MAC address table, which is generated by a Control Plane protocol to forward traffic.

The Control Plane provides information to the Data Plane about where to forward the data. An example of this is a routing protocol that calculates the best path to the destination and provides the outgoing interface for network traffic.

The devices can be managed remotely by network management software (via SNMP) in the Management Plane. Protocols at the Management Plane include Telnet, SSH, and FTP.

Separating the Control Plane from the Data Plane is one solution for programmable networks to control the network. Instead of managing devices one by one, a centralized controller influences the traffic of all devices on the network. Decoupling the Control Plane and Data Plane introduces a new architecture called software-defined networking or SDN. SDN is an architecture where the Control Plane and Data Plane are separated

REST - RESTful API design is designed to leverage existing protocols. Although REST can be used over many protocols, it usually takes advantage of HTTP when used for Web APIs. To be considered as REST-based API, it must follow an architectural style of development that consists of rules/constraints. There are six key constraints to the REST API design to be aware of when deciding whether this is the right API type for your project.

1. Client-Server
2. Stateless
3. Caching
4. Uniform Interface
5. Layered System
6. Code on Demand (optional)

JSON data is written as name/value pairs. A name/value pair consists of a field name (in double quotes), followed by a colon, followed by a value:

```
"name":"Alan"
```

JSON can use arrays. Array values must be of type string, number, object, array, boolean or null. For example:

```
{
"name":"Alan",
"age":50,
"jobs":[ "Admin", "Dentist", "Carpenter" ]
}
```

Configuration Management tools include SaltStack, Ansible, ~~Puppet, and Chef~~ (among others), and were created to install and manage software on existing server instances (as well as compatible switches and routers). They can perform tasks such as the installation of packages, scripts, or configuration files, and the initialization of services.

Terraform is an open-source infrastructure as code (IaC) tool developed by HashiCorp that enables developers and operations teams to define, provision, and manage cloud and on-premises infrastructure using declarative configuration files written in HashiCorp Configuration Language (HCL)

AI

Machine Learning (ML) in network operations is a subset of AI that enables systems to learn and improve from data patterns without explicit programming, facilitating tasks like anomaly detection and traffic forecasting in Cisco environments.

Predictive AI in network operations uses ML algorithms to analyze historical and real-time data for forecasting future events, such as potential outages or bandwidth needs, allowing proactive maintenance and optimization.

Generative AI in network operations leverages ML models to create new synthetic data or configurations, such as simulated network scenarios or automated scripts, enhancing design testing and troubleshooting efficiency for Cisco networks.

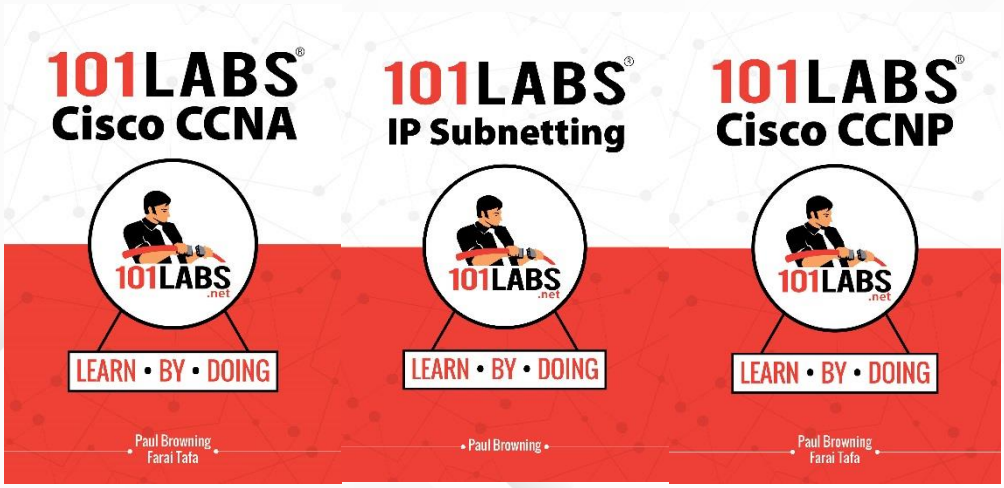
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